# SAINT MICHAEL COLLEGE OF CARAGA

**ONLINE REGISTRATION SYSTEM FOR THE CIVIL REGISTRAR'S OFFICE OF BUENAVISTA AGUSAN DEL NORTE**

A Thesis Presented To
The Faculty of
College of Computing and Information Sciences

In Partial Requirements for the Degree
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

By

JOUSELO REY G. NAKILA
ALEX ANDREA B. CABAN
GERLY GRACE P. OMPAD
RICHARD JOHN L. BUDLAT
VINCE CARTER O. YAMBA

ADVISER: REA MIE A. OMAS-AS

March 2026

TABLE OF CONTENTS


| | | Pages |
|---|---|---|
| TITLE PAGE | | i |
| CERTIFICATE OF RESEARCH APPROVAL | | ii |
| ABSTRACT | | iii |
| DEDICATION | | iv |
| ACKNOWLEDGEMENT | | v |
| TABLE OF CONTENTS | | vi |
| LIST OF TABLES | | viii |
| LIST OF FIGURES | | x |
| CHAPTER | | |
| 1 | **INTRODUCTION** | 1 |
| | 1.1 Project Context | 1 |
| | 1.2 Review Related Literature | 5 |
| | 1.3 Objective of the Study | 12 |
| | 1.4 Scope and Limitations | 12 |
| | 1.5 Definition of Terms | 13 |
| 2 | **TECHNICAL BACKGROUND** | 15 |
| | 2.1 System Architecture | 15 |
| | 2.2 Conceptual Diagram | 17 |
| | 2.3 Use Case Diagram | 18 |
| | 2.4 Activity Diagram | |
| | 2.5 Sequence Diagram | 19 |
| | 2.6 ERD (Database Design) | 22 |
| | 2.7 User Interface Design (Prototype) | 23 |
| | 2.8 Software Platforms, Development Environments and Tools | 40 |
| | 2.9 Hardware Requirements | 44 |
| 3 | **SOFTWARE DEVELOPMENT AND TESTING** | 45 |
| | 3.0 Development Process | |
| | 3.1 Testing Process | |
| | - Accuracy Test | |
| | - Functionality Test | |
| | - Performance Test | |
| 4 | **SUMMARY, CONCLUSIONS AND RECOMMENDATIONS** | 46 |
| | 4.1 Summary of Findings | |
| | 4.2 Conclusion | |
| | 4.3 Recommendations | |
| REFERENCES | | 50 |

CURRICULUM VITAE 59
APPENDICES
CERTIFICATE OF TECHNOLOGY BASED ASSESSMENT 68
PUBLISHABLE JOURNAL 100

**SAINT MICHAEL COLLEGE OF CARAGA**
College Of Computing and Information Sciences

**CERTIFICATE OF RESEARCH APPROVAL**

This thesis entitled **"**Click here to enter text.**"** presented and submitted by Click here to enter text. in partial fulfillment of the requirements for the degree of Click here to enter text. is hereby accepted and recommended for Oral Examination.

**DELETE BOX AFTER READING**

**Kindly, fill out the names of the panelists in your title hearing, research proposal defense, and final defense.**

**Make sure to get their names and their titles and put it in capital letters**

**Example:**

**KENNETH IAN B. BARRERA, MA**

Click here to enter text.
Adviser

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Chairperson

ACCEPTED as partial fulfillment of the requirements for the degree of Click here to enter text.

APPROVED by the tribunal at the Oral Examination with the grade of _____.

Click here to enter text.
Adviser

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Member

Click here to enter text.
Chairperson

**COLLEGE OF COMPUTING AND INFORMATION SCIENCES**

*ABSTRACT*

| | |
|---|---|
| **TITLE:** | XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX |
| **AUTHOR:** | XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX |
| **DEGREE:** | XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX |
| **ADVISER:** | XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX |
| **PLACE OF PUBLICATION:** | XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX |
| **DATE** | XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX |
| **PAGES** | XXXXXXXXXXXXXXXX XXXXXXXXXXXXXXXX |

***(RESEARCH DESIGN)***

**I. OBJECTIVES**

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

**II. METHODOLOGY**

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

**III. FINDINGS**

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

**IV. RECOMMENDATIONS**

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

**KEYWORDS:** XXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

**gDELETE BOX AFTER READING**

**RESEARCH DESIGN -refers To The Specific Approach Use In The Research E.x. DESCRIPTIVE QUALITATIVE**

**Objectives**
- **Write a short introduction of your paper. Provide the direction of the study and nuances.**

**Methodology**
- **Provide the specific method that was used in your paper.**

**Findings**
- **Highlight the significant findings of your paper.**

**Recommendation**
- **Just highlight the primary recommendation**

**KEYWORDS: PROVIDE AT LEAST 5**

**Check guidelines for writing an abstract for further details. Max of 250 words from objective 1-5**

## DEDICATION

This study is dedicated to our beloved families, whose unconditional love, patience, and unwavering support have been our constant source of strength and inspiration throughout this journey.

To our friends, advisers, and especially to our parents, who believed in our dreams and encouraged us to pursue our goals despite the challenges, your sacrifices and guidance motivated us to persevere and give our best.

To our research adviser, Ms. Rea Mie A. Omas-as, thank you for your invaluable guidance, encouragement, and dedication, which helped us overcome difficulties and complete this study successfully.

To our friends and classmates, your support, understanding, and companionship made this research journey meaningful and memorable.

This work would not have been possible without all of you.

## ACKNOWLEDGEMENTS

We, the researchers, would like to express our sincere gratitude to everyone who helped us to finish our study, "Online Transaction System of the Civil Registrar's Office of Buenavista, Agusan del Norte."

We, the researchers, would like to express our sincere gratitude to Ms. Rea Mie A. Omas-as, our research adviser, for her unwavering support, encouragement, and knowledge, all of which were crucial to the completion of our study. We also want to sincerely thank our professors and panelists for their insightful comments and helpful criticism, which enabled us to enhance and perfect our study.

We, the researchers also acknowledge the Buenavista, Agusan del Norte, Civil Registrar's Office for their assistance and collaboration. In order to finish this system, your assistance and willingness to provide the necessary information were vital in completing this system.

To our beloved families, thank you for your unwavering love, understanding, and financial support. To our friends, classmates, and advisers, your encouragement and companionship throughout this journey motivated us to persevere and complete this research successfully.

Special thanks to Teresita K. Valdehueza, Head of the LCR staff, and Joanne A. Pal and Naomie A. Ayala of the LCR faculty, for their valuable cooperation and technical assistance during the system development process, all of individuals, and team members for their technical contributions and assistance during the system development process.

Above all, we, the researchers, extend our profound gratitude to everyone who, in one way or another, has shared their time, knowledge, and support. Each contribution big or small became an essential part of this study's success. This study would not have been possible without your guidance, help, and unwavering support.

## LIST OF TABLES


| Table | | Pages |
|---|---|---|
| 1 | Software Platform, Development Environments, and Tools | 47 |
| 2 | Hardware Requirements | 49 |
| 3 | Functional Suitability | 54 |
| 4 | Performance Efficiency | 54 |
| 5 | Usability | 55 |
| 6 | Respondent's Distribution | 55 |

## LIST OF FIGURES


| Figure | | ~~Pages~~ |
|---|---|---|
| 1 | Agile Methodology | ~~16~~ |
| 2 | System Architecture | ~~17~~ |
| 3 | Conceptual Diagram | ~~18~~ |
| 4 | Use Case Diagram | ~~20~~ |
| 5 | Activity Diagram LCR Staff | ~~21~~ |
| 6 | Activity Diagram Citizen/Applicant | ~~22~~ |
| 7 | Activity Diagram Admin/Programmer | ~~23~~ |
| 8 | LCR Staff Sequence Diagram | ~~24~~ |
| 9 | Applicant/Citizen Sequence Diagram | ~~25~~ |
| 10 | Admin/Programmer Sequence Diagram | ~~26~~ |
| 11 | Entity Relationship Diagram (ERD) | ~~27~~ |
| 12 | Landing Page Dashboard | ~~28~~ |
| 13 | Citizen Login Page | ~~29~~ |
| 14 | LCR Staff Login Page | ~~30~~ |
| 15 | Account Creation Page | ~~30~~ |
| 16 | Citizen Portal of the Local Registry System | ~~31~~ |
| 17 | Birth Registration Requirements Page | ~~32~~ |
| 18 | Death Registration Requirements Page | ~~33~~ |
| 19 | Marriage Registration Requirements Page | ~~34~~ |
| 20 | Registration Requirements Page | ~~35~~ |
| 21 | File Upload Section Registration Requirements | ~~36~~ |
| 22 | Certificate of Live Birth Form | ~~37~~ |




| | | |
|---|---|---|
| 23 | Application for Marriage License Form | ~~38~~ |
| 24 | Certificate of Death Form | ~~39~~ |
| 25 | Civil Registry Staff Dashboard | ~~40~~ |
| 26 | Statistics Dashboard | ~~41~~ |
| 27 | Timely Birth Records Marital Table | ~~41~~ |
| 27.1 | Timely Birth Records Non-Marital Table | ~~42~~ |
| 27.2 | Birth Records Delay Table | ~~42~~ |
| 28 | Timely Death Records Table | ~~42~~ |
| 28.1 | Delay Death Records Table | ~~43~~ |
| 29 | Marriage Records Table | ~~42~~ |
| 30 | Birth Registration Validation Interface | ~~42~~ |
| 30.1 | Birth Registration QR code | ~~43~~ |
| 31 | Death Registration Validation Interface | ~~43~~ |
| 31.1 | Death Registration QR code | ~~43~~ |
| 32 | Marriage Registration Validation Interface | ~~44~~ |
| 32.2 | Marriage Registration QR code | ~~44~~ |
| 33 | Uploading Requirements Interface | ~~44~~ |
| 34 | Certificate of Live Birth Forms Interface | ~~45~~ |

# CHAPTER 1

# INTRODUCTION

## 1.1 Project Context

Efficient and accessible civil registration systems are essential for accurately recording vital events like births, deaths, and marriages. According to the United Nations, strong civil registration and vital statistics (CRVS) systems are key for good governance, public health, and sustainable development [1]. However, paper-based methods often cause delays and limited accessibility, making it difficult for individuals to verify their legal identity and access essential services [2]. As governments transition to digital platforms, more countries recognize the value of e-government tools in enhancing transparency, accuracy, and citizen services.

The COVID-19 pandemic revealed both the strengths and weaknesses of global e-governance systems. Digital platforms definitely made things easier to access and more transparent. However, honestly, many government offices are still stuck in the Stone Age with their clunky systems and tech that belongs in a museum[3]. Several issues, including slow and inefficient paper-based processes, lengthy waiting times, manual form errors, and limited accessibility for citizens in rural areas, hinder the acquisition of Civil Registry Documents. These problems result in low user satisfaction, wasted time and resources, and contribute to environmental concerns due to high paper consumption and waste generation[4]. Both studies underline the urgent need for effective and transparent digital governance. While Villena et al.[4] Focus on local digitalization efforts in the Philippines, emphasizing efficiency and environmental sustainability, as noted by Naqvi et al.[3] Expand the perspective to global post-pandemic challenges, stressing collaboration, data design, and cloud-based adaptability. Together, they highlight a continuous progression of e-governance from basic digitization of services to developing robust, integrated, and crisis-ready governance systems. Tapangan [5] highlighted how digitalizing civil registration in the Philippines

significantly aided them in managing their operations during the pandemic. Villena et al.[4] essentially stated that making this information available online makes it significantly easier for everyone, especially those living in remote areas. Both studies highlight old problems: shaky infrastructure, wonky resource management, and inconsistent rules enforcement. So yeah, good ideas, but plenty of roadblocks.

On the international scale, Angin et al.[6] studied how Indonesia transformed its population and civil registration services during the COVID-19 pandemic by introducing the SILAO system. This system aimed to make registration contactless and efficient, but the study found several problems. Many staff and citizens struggled with the system due to its difficult navigation. The SILAO platform also required a strong internet connection, posing a challenge for users in remote areas. Some steps still had to be done in person at registration offices. Limited digital skills among employees and the public also slowed adoption. While SILAO was a step forward, it did not fully deliver an accessible and smooth digital service for everyone.

At the local level, Celeste et al. [7] assessed how extensively citizens in a municipality in Negros Occidental registered simple events such as births, marriages, and deaths. The findings indicated that birth and marriage registrations were typically well-managed, but death registrations were often not attended to, suggesting a lack of perceived value among the population. The study highlighted some challenges that rendered registration less effective, including unclear requirements, far-off office locations, and the absence of directions, particularly for rural and poor populations. Although the authors called for electronic systems to enhance the Civil Registry and Vital Statistics (CVRS), they failed to investigate how digital tools would simplify the process. Individuals continued to register in person with paper forms, with no website or notifications. These problems indicate a demand for solutions that are simple to use digitally. For this, the present study proposes a paperless registration system with email notifications and an

easy interface to make it more accessible, efficient, and responsive to community needs.

The present study proposes a digital system with real-time tracking, online appointment scheduling, and automated reports to address these persistent challenges. The system will be developed based on user needs, stakeholder feedback, and repeated testing to ensure it is easy to use and accurate. Key features will include role-based access, secure data backup, and notifications by SMS or email to keep users informed and ensure transparency. The system aims to reduce paperwork, avoid duplicate work, and improve accountability in the Civil Registrar's Office by moving document submission, checking, and storage online.

This study is critical because it helps advance local e-governance and digital public services. The system provides a lasting solution to enhance legal identity management and supports the Philippine government's initiative toward paperless, citizen-centric transactions. Kumar et al. [13] noted that digital tools in public offices build trust by making processes more transparent and efficient, while Vicente et al. [8] found that digital platforms make citizens happier by speeding up access and cutting wait times. This study addresses problems in the current manual process and shows how digital changes can improve local governance and encourage other towns to try similar solutions.

The new Online Transaction System for the Civil Registrar Office of Buenavista, Agusan del Norte, aims to improve how the municipality handles civil registration. Currently, the office uses paper forms for birth, marriage, and death certificates, which often causes slow service, lost or repeated records, and long waits for citizens. These problems delay document release and make it hard for staff and clients to track requests. The new system lets users submit documents online, check their application status, and access records in a safer and more organized way. Compared to Indonesia's SILAO system, which lacks user-friendly features such as automated email notifications [6], and the conventional paper-based registration procedure previously used in Negros Occidental

[7], this new system aims to simplify life. The proposed system provides users with automatic email notifications and features a user-friendly interface accessible to users with limited computer experience. It is a significant upgrade from the previous manual handling of records, as it saves people from spending excessive time traveling back and forth to multiple offices to obtain documentation. The digital method also reduces employees' workload by avoiding excessive paperwork and data management. Moreover, it also enables local authorities to obtain accurate and real-time information, replacing outdated and unorganized files. In sum, adopting this system significantly improves the efficiency, transparency, and usability of civil registration procedures in Buenavista.

## REVIEW OF RELATED LITERATURE

This chapter outlines the relevant literature supporting the researchers' study of the Online Transaction System for the Civil Registrar's Office in Buenavista, Agusan del Norte. It highlights the problems in traditional registration methods, the benefits of computerized substitutes, and the need for user-centric design principles. Various studies, articles, journals, and other web-based papers discuss computerization in civil registration centers.

### Government Civil Registration System

According to Naqvi et al.[3], The COVID-19 pandemic exposed both the advantages and the disadvantages of the world's e-governance systems. Since online governance enhanced transparency, access, and civil involvement, the organizations responsible for these systems poorly set them up to meet the sudden surge in demand. The authors highlighted three important factors for resilience: collaboration across government sectors, strong data management, and cloud-based services. Villena et al.[4], The transition to paperless e-government in the Philippines responds to citizens' requirements for faster, more transparent, and more accessible civil registration services. Their results were that online channels assisted in viewing such records as birth, marriage, and death certificates. Nonetheless, some deterrents remain, including low digital literacy for some segments of society, limited access in some regions, and common mistrust of online systems. It hinders adoption and creates gaps in inclusiveness.

Vicente et al.[8], emphasized that the digitalization of Peru's civil registration system (RENIEC) brought mixed results. According to the study, digital birth and divorce certificates typically raised user satisfaction. In contrast, digital death certificates frequently led to dissatisfaction among senior users who were not very at ease with the digital procedures. It suggests that even though digitalization simplifies matters for services and restricts paperwork, it is still a function of how

appropriate the system is to individual needs and abilities at different ages. The following studies illustrate the virtues and vices of taking services for civil records and governance to the online sphere. As Naqvi et al.[1] As mentioned, there should be firm and flexible systems. On the other hand, Santos et al. were concerned with citizens' preparedness and infrastructure support. Additionally, Vicente et al.[8] highlighted the roles of service design and population components in user satisfaction. In total, these thought processes mean that digitalized governance is not merely about digitizing service delivery, but about making sure that the systems are resilient, accessible, and responsive to the extensive needs of citizens. In our case study, successful e-government must integrate contemporary technology with infrastructure investment, promote citizens' awareness, and plan services with user diversity in mind.

**Digital Transformation of Civil Registry Systems**

According to Tapangan[5], the City Civil Registry Office in Cagayan de Oro, Philippines, uses digital technologies. Tapangan[5] further mentioned that these digital technologies help to enhance the efficiency and effectiveness of its operations, like the City Civil Registry Information System (CCRIS), digitized records, central database, and online applications. They replace the old manual ways, making records more accurate. Tapangan[5] also says this is more reliable. But Angin et al.[6] argue that using new technology after crises is hard. They require sturdy equipment, adequate infrastructure, trained staff, and efficient processes. Interestingly, Ndiyun et al.[2] argue that sustainability depends on addressing access and capacity gaps. For the current investigation, these lessons emphasize that digital systems must be sustainable, inclusive, and citizen-centric so that civil registration can support long-term development aspirations and unforeseeable challenges. More resilient. Those initiatives solved ubiquitous issues like time-consuming manual filing, disaster damage vulnerability, and extended office queuing. They also improved accuracy,

reduced costs, and made services more accessible for citizens, especially during the COVID-19 pandemic.

According to Lestari et al.[9], Civil registration and vital statistics (CRVS) systems are the most reliable data source for monitoring health-related Sustainable Development Goals (SDGs), particularly when linked with national identification systems. Researchers could link records generated from civil registration, health, and migration data if each citizen could be assigned a unique identification number at birth. It will provide more reliable data for governments to use for planning and resource allocation, making it easier for citizens to access many services, including healthcare, education, and social protection.

According to Shoewu et al.[10], Nigeria's birth/death system uses manual record generation, storage, and retrieval methods. That has resulted in error-prone, inefficient processes and the tendency to lose vital data, thus rendering population data and planning ineffective. The study authors created an online system to automate record-keeping, eliminate paperwork, and build a more robust information base for decision-making. This research highlighted that digital solutions are key to preserving rights, delivering services, and contributing to demographic planning. Researchers have to designed an online system that would maintain records in automatic mode, weeding out the paperwork and at the same time more accurate database for making any decision. The study emphasized that a digital system is crucial to safeguarding people's rights, adding value to services, and demographic planning.

According to Angin et al.[6], The COVID-19 pandemic pushed the digital transformation of public services in Indonesia, especially in civil registration. The Situbondo Civil Registry Office has introduced the SILAO online system, enabling citizens to apply for ID cards, family cards, and certificates from their homes. However, internet and staff knowledge issues still result in an

imperfect system. Ndiyun et al.[2] Cameroon made a significant change in 2024 by digitizing its civil registration system. Changes included electronic declaration and a database, making certificates electronic and improving access and transparency. This improved service delivery; however, there are still issues with weak internet access, low digital literacy, and inadequate infrastructure. The authors emphasized that policymakers and educators must address such barriers to achieve long-term effectiveness through improved policies and citizen education. They indicate that online conversion of civil registration can highly enhance efficiency, transparency, resilience, and service access. However, the correct support for systems depends on the infrastructure, inclusive policy, and citizens' preparedness.

**Assessment of Civil Registration Awareness**

According to Celeste et al.[7], Civil registration in the Philippines shows that most residents know and comply with registration requirements, especially for births and marriages. However, death registration remains less consistent. Their study also found that women and people with higher levels of education tend to comply more. Even with these efforts, obstacles like the unavailability of valid identification cards, transportation costs, and mix-up with documentary requirements still hinder some citizens from registering. In a study by Fruchtman et al. [11], the COVID-19 pandemic revealed how essential Civil Registration and Vital Statistics (CRVS) systems are in supporting public health and legal identity protection. Their research emphasized that resilient CRVS systems, capable of continuing operations during crises, adapting to disruptions, and protecting human rights, are critical for effective governance. The authors introduced the framework for Resilient CRVS Systems, which identifies seven key capacities that enable countries to prepare for, respond to, and recover from emergencies. Findings from case studies in countries such as Guatemala, Ecuador, and Fiji showed that nations with digitized, decentralized, and well-integrated CRVS systems could better maintain services and monitor mortality data during the

pandemic. The study recommends governments invest in digitalization, interoperability, decentralization, and inclusivity to strengthen CRVS resilience and ensure equitable access to civil registration services.

As defined by Mikkelsen et al.[12], CRVS systems play two important roles: giving legal identity through birth registration and providing vital fertility data to governments. Although global initiatives have focused on expanding birth registration towards attaining Sustainable Development Goal (SDG) 16.9, the research highlighted that analysts have not yet fully utilized the value of CRVS systems in gathering high-quality fertility data. If utilized effectively, good birth data can inform population projections and assist governments in planning for schools, clinics, and hospitals. Separately, the works establish that civil registration keeps improving but remains bedeviled by issues that take away from the effectiveness of civil registration as a complete entity. As noted by Celeste et al.[7], It is crucial to address the issues that individuals encounter, including cost and the requirement for official identification. Fruchtman et al. [11] emphasize that Governments must strengthen Civil Registration and Vital Statistics (CRVS) systems through digitalization, interoperability, and good governance to remain functional and resilient during emergencies such as the COVID-19 pandemic. This study's results indicate that a strong civil registration system includes solutions targeting citizens and institutional improvement, and maximizes the use of registration data to support country development.

**Secure and Efficient Management of Vital Records and Population Data**

Sudan Kumar Mulmi [13] states that the vital registration system in Nepal has been improving over the years, with an increase in the number of people registering births, deaths, marriages, divorces, and internal migrations. Nonetheless, the lack of awareness about the importance of registration remains a reason for many to delay it, while outdated laws also slow down the process. The study pointed out that raising public awareness and updating legal provisions are necessary

to make the system more reliable and efficient. According to Herlinda et al.[14], The COVID-19 pandemic encouraged digital services in Indonesia's population administration, such as the SIPENDUDUK online system in Pekanbaru. This platform made services more accessible, but faced problems like server breakdowns, limited processing capacity, and low public awareness. The SIPENDUDUK platform, on the one hand, has brought the population administration service closer to the public, allowing them to make submissions from home. However, the system still faces several fundamental implementation challenges, such as frequent server breakdowns and limited data processing capacity, while using SIPENDUDUK. The authors stressed that technology alone is not enough; The SIPENDUDUK must promote proper preparation, and educators must actively use public education to make digital systems effective.

According to Sumirat et al.[15], Indonesia introduced absolute liability statements (SPTJM) to help families from unregistered marriages get critical population documents like birth certificates. While this made access easier, the study raised concerns about legal certainty, especially regarding inheritance rights. The authors explained that although SPTJM helps in the short term, it cannot replace formal legal processes such as court confirmation of marriages. Sordo et al.[16], say that late registration of deaths makes it hard to track deaths during health emergencies like the COVID-19 pandemic. They developed a statistical model to obtain nearly all the weekly death counts from incomplete data. It provides less biased, timelier information and allows governments to respond better to crises.

Nalband et al.[17] Manual civil registration systems using paper-based systems are inefficient and suffer from many issues, such as record forgery, loss, and untimeliness. They developed a digital civil registration system using ASP.NET and SQL Server to overcome the challenges. The system performed the tasks of data input, document verification, and certificate generation, while at the same time, it enhanced security through encryption and QR codes. The various studies

highlight that civil registration has improved, but issues still need to be addressed. Sudan Kumar Mulmi [13] stated that we must update laws and educate about the civil registration process, and Herlinda et al.[14], stated that digital systems need technology and community engagement to work correctly. Sumirat et al.[15], pointed out the paradox between facilitating access to services and legally protecting these services. Sordo et al. [16] showed how new methodologies can improve the time-lag in data reporting. The main point was that Nalband et al.[17], demonstrated that electronic platforms could replace the traditional paper systems in civil registration, gaining better accuracy and security. The researchers' heads-up indicated that civil registration modernization must consider not only hardware improvements but also legal reforms, public consciousness, and access and dependability strategies.

**Vital Statistics: A User-Friendly System for Efficiency and Sustainability**

According to Vicente et al.[8], The move toward digital civil registration in Peru, managed by RENIEC, has improved accessibility and efficiency. Their study showed that digital birth and divorce registration platforms increased user satisfaction. In contrast, death registration had mixed results because older users found it challenging to adapt to online systems. In Indonesia, Anityasari et al.[18], looked at the transition of Surabaya's civil registration services into digital platforms like KLAMPID COVID. They found online systems sped up processes, reduced paper usage, and helped with health regulations during the pandemic. However, it also increased electricity and internet usage.

In a similar way, Herdianti et al.[19], worked on a mobile web-based system in Binjai City because registration took too long, and the line was always long. The new system made things smoother, with fewer errors, and users liked it a lot. The research shows that when applied in civil registration, digital technology speeds up the entire process and makes it user-friendly; therefore,

trust is built. They highlight that technology alone is not enough to solve the issues in civil registration. Decision-makers must consider practical barriers such as costs and access. The present study suggests that civil registration reforms combine technology with user-centered design and supportive policies to ensure that services are efficient but also inclusive and sustainable.

**Web-Based Application System**

According to Kaewsuwan et al. [28], creating a web-based registration system was to solve the problems with manually enrolling students at Pitchayabundit College in Thailand. Their research utilized a digital platform to streamline registration, record-keeping, and grade submission tasks. The system reduced the likelihood of mistakes, accelerated processing, and increased user satisfaction. The researchers said that using information technology in administrative work makes data more accurate, easier to find, and more efficient overall. It shows how going digital can make old ways of keeping records more up-to-date.

According to Celeste et al. [29], civil registration constitutes the legal documentation of significant events, including births, marriages, and deaths, as stipulated by Republic Act No. 3753, recognized as the Civil Registry Law of the Philippines. Their research examined the levels of awareness, compliance, and challenges encountered by residents of Negros Occidental municipalities concerning civil registration. The results showed that people were very aware of and followed the rules for registering births and marriages, but not so much for registering deaths. The study demonstrated that education, gender, lack of identification, and limited access to places influenced the results. It highlighted that increasing awareness of and access to services is crucial for encouraging more people to participate in government documentation processes.

According to Vijay Panwar [30], digital civil registration and legal identity systems are vital in promoting inclusivity and accountability in governance. The report argued that traditional paper-based registries are inadequate in addressing the needs of modern society, as they often exclude marginalized groups from legal recognition and essential public services. By shifting to digital civil registration and vital statistics (CRVS) systems, countries can enhance data accuracy, accessibility, and timeliness.

The study emphasized that integrating civil registration with identity management and other government systems will ensure everyone participates. It aligns with Sustainable Development Goal (SDG) 16.9, which promotes "legal identity for all by 2030." *The* reviewed studies collectively emphasize the transformative impact of digital systems in improving administrative processes, service accessibility, and data management. Kaewsuwan et al.[28] demonstrated how web-based applications streamline operations and reduce inefficiencies in institutional workflows. Similarly, Celeste et al. [29] highlighted the need for accessible and user-friendly systems to encourage citizen participation in civil documentation. Vijay Panwar [30] expanded this perspective by underscoring the importance of digital civil registration systems in ensuring inclusivity and accurate data management for effective governance. Together, these studies support the current research's aim to develop an online documentation requisition system that promotes efficiency, transparency, and accessibility for local citizens.

### 1.3 Objective of the Study

The researchers' study aims to design and develop an online transaction system for the Civil Registrar's Office of Buenavista, Agusan del Norte, to streamline civil registration processes and enhance service delivery for citizens.

Specifically, it aims to:

1. Digitalize and streamline the Local Civil Registry (LCR) transactions of the Municipality of Buenavista, particularly focusing on birth, marriage, and death registration services to ensure faster and more accurate processing.
2. Design an online system that allows applicants to upload necessary documents, track their application status in real time, and receive updates through email notifications.
3. Establish ethical standards in data management by implementing encryption or hashing techniques that ensure the confidentiality, integrity, and security of sensitive information, thereby protecting user privacy and preventing unauthorized access or data misuse.
4. Evaluate the developed system based on the ISO 25010 Software Quality Standards to assess its functionality, efficiency, and overall usability.

### 1.4 Scope and Limitations

This capstone project seeks to design and develop a web-based Online Transaction System for the Civil Registrar's Office of Buenavista to replace manual processes for civil registration services such as birth, marriage, and death registrations. The system is only for the bona fide residents of Buenavista who present proper identification. The users can submit applications, track their status, and receive notifications through the online system, while designated personnel from

the Civil Registrar shall handle transactions, evaluate applications, and generate reports through a standard administrator account.

The system will operate exclusively in the Civil Registrar's Office, but users can also access it through computers and cell phones. It supports online access to provide clients with convenient usage anytime and anywhere, as long as an internet connection exists. The system will not support online payment since the Treasurer's Office will process all payments separately.

**1.5. Definition of Terms**

**Agile Methodology** – the methodology adopted by the researchers in constructing the system in iterative cycles that facilitated continuous prototyping, assessment, and refinement.

**Civil Registry Documents** – the authorized documents from the Local Civil Registry (LCR), such as birth, marriage, and death certificates.

**Real-Time Tracking** – the system features that allow users to monitor the current status of their online submissions. The users are automatically notified immediately by SMS or Email whenever their application advances to the next level (e.g., received, under review, approved), thus improving transparency and helping to gain confidence in the process.

**Role-Based Access Control (RBAC)** – the security measures in the system to control access and features based on each user's role.

# CHAPTER 2

## TECHNICAL BACKGROUND

### Research Design

This study will use a Design and Development Research (DDR) approach, which involves creating and testing the Online Civil Registry System. This technology-based solution enhances record management, minimizes processing delays, and increases transparency within civil registration procedures.

### Software Development Life Cycle




*Figure 1. Agile Methodology*

Figure 1 illustrates the use of the Agile Methodology Software Development Life Cycle in developing the Online Civil Registry System. Agile methodology uses short, repeated sprints to design, build, and test functional parts. This method allows for flexibility and encourages collaboration among LCR Staff instead of following a set order. The first step is to plan and gather requirements. Most important features, like user logging, usage tracking, and reporting. Each sprint includes the design, development, and testing of separate modules, making it easy to use

feedback to improve the system's efficiency. This Agile method ensures the system is user-friendly, flexible, and capable of handling the evolving needs of civil registration services.

**2.1 System Architecture**

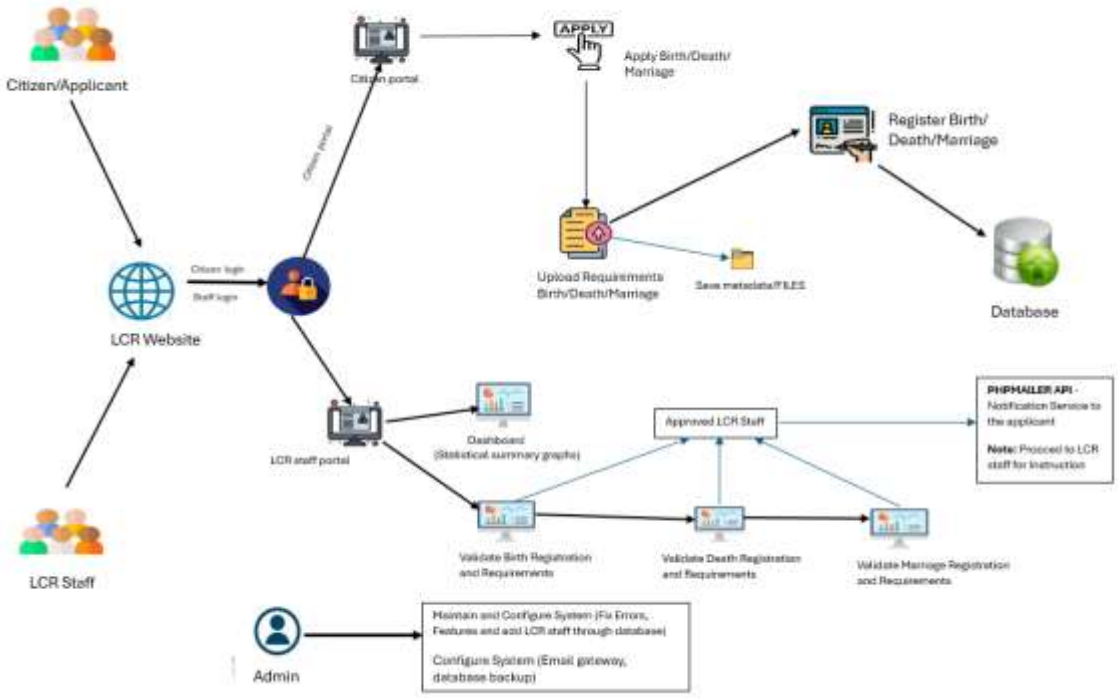



*Figure 2: System Architecture*

Figure 2 presents the three-tier architecture of the Online Transaction System, comprising the Client, Application, and Data Tiers. The Client Tier delivers the user interface, developed with HTML, CSS, and JavaScript, and supports access via web browsers. The Application Tier executes business logic using PHP, manages authentication and role-based access control, handles file storage and notifications, and produces reports including graphical representations and PDF exports.

## 2.2 Conceptual Diagram

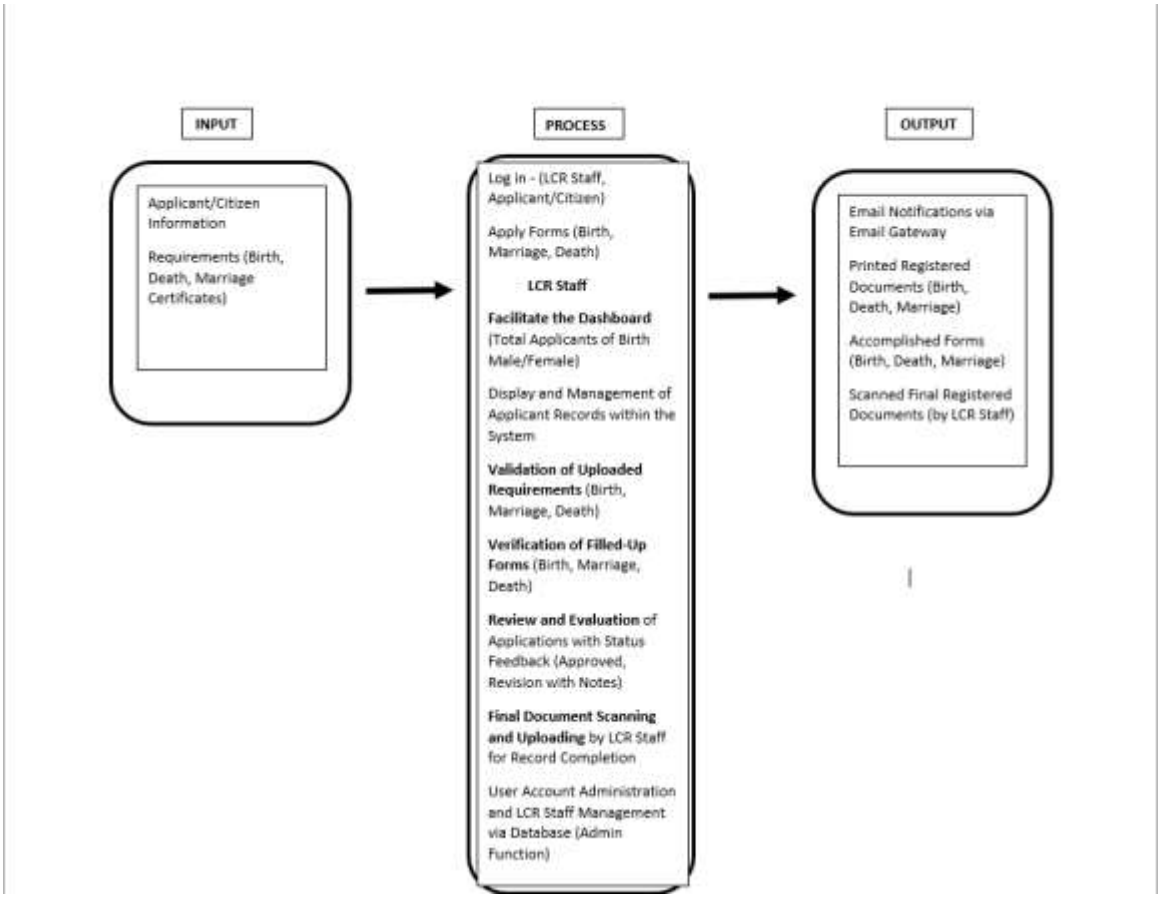



*Figure 3: Conceptual Diagram*

Figure 3 illustrates the conceptual diagram of the Local Civil Registry Office's (LCR) Online Transaction system. In addition to creating profiles and logging in, applicants can upload necessary documents, register significant life events such as births, marriages, and deaths, and track the progress of their applications. Applicants receive email notifications or text messages to keep them updated on the status of their applications. LCR staff review and check applications, review the uploaded documents, and decide whether to approve, deny, or ask for changes. Staff members also write reports that examine the births that staff have registered. The administrator or

programmer keeps the system running by setting up features, fixing errors, and running the system.

### 2.3 Use Case Diagram

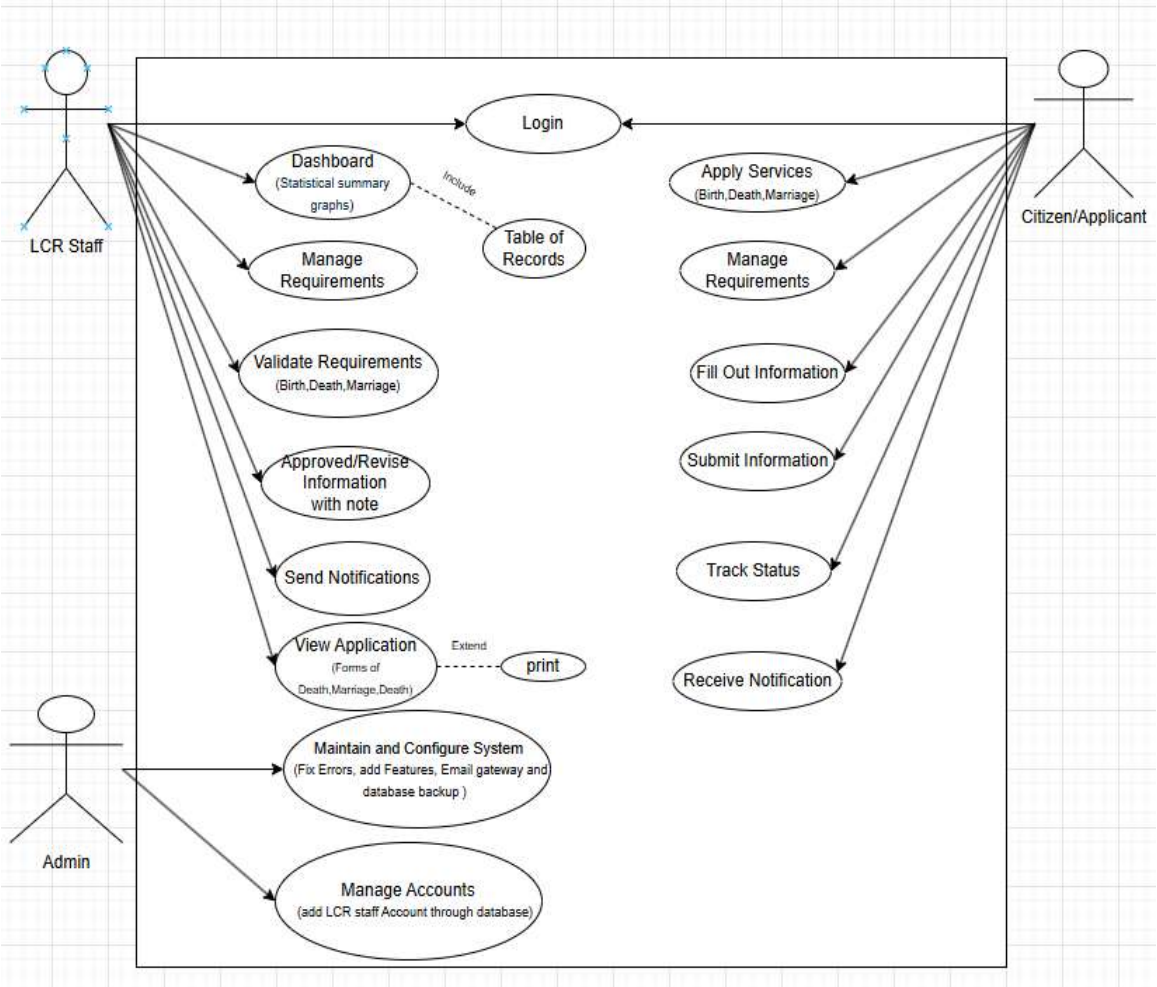



*Figure 4: Use Case Diagram*

Figure 4 shows the use case diagram of The Local Civil Registry Office's (LCR) Online Transaction System. In addition to creating profiles and logging in, applicants can upload necessary documents, register important events like births, marriages, and deaths, and track the progress of their applications. The progress will notify applicants via email or SMS. LCR staff evaluate applications, verify supporting documents, and decide on approval, rejection, or the need for

revisions. System stability relies on feature configuration, technical issue resolution, and operational management by the admin or programmer.

**2.4 Activity Diagram**

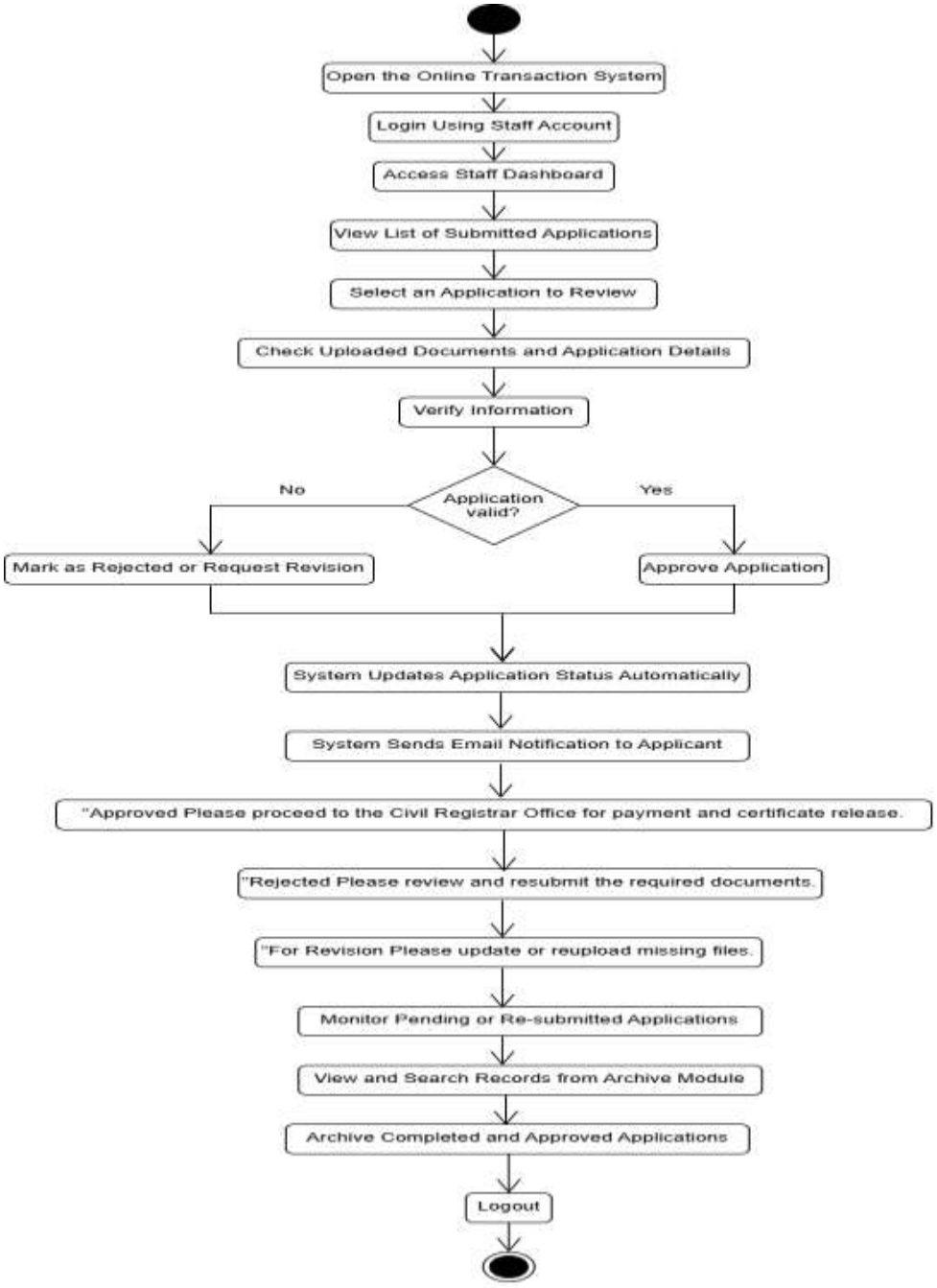



*Figure 5: Activity Diagram LCR Staff*

Figure 5 illustrates the activity diagram workflow of the Local Civil Registry (LCR) Staff as they manage and handle online civil registration applications. The process starts when staff members log in with their staff account and open the online civil transaction system. Once on the staff dashboard, they can view submitted applications and select one to review. The staff reviews the uploaded documents and application information to ensure everything is correct. The staff may reject the application or request a revision if it is invalid. If valid, the staff approves it. The system will automatically update the application status and send an email notification to the applicant.

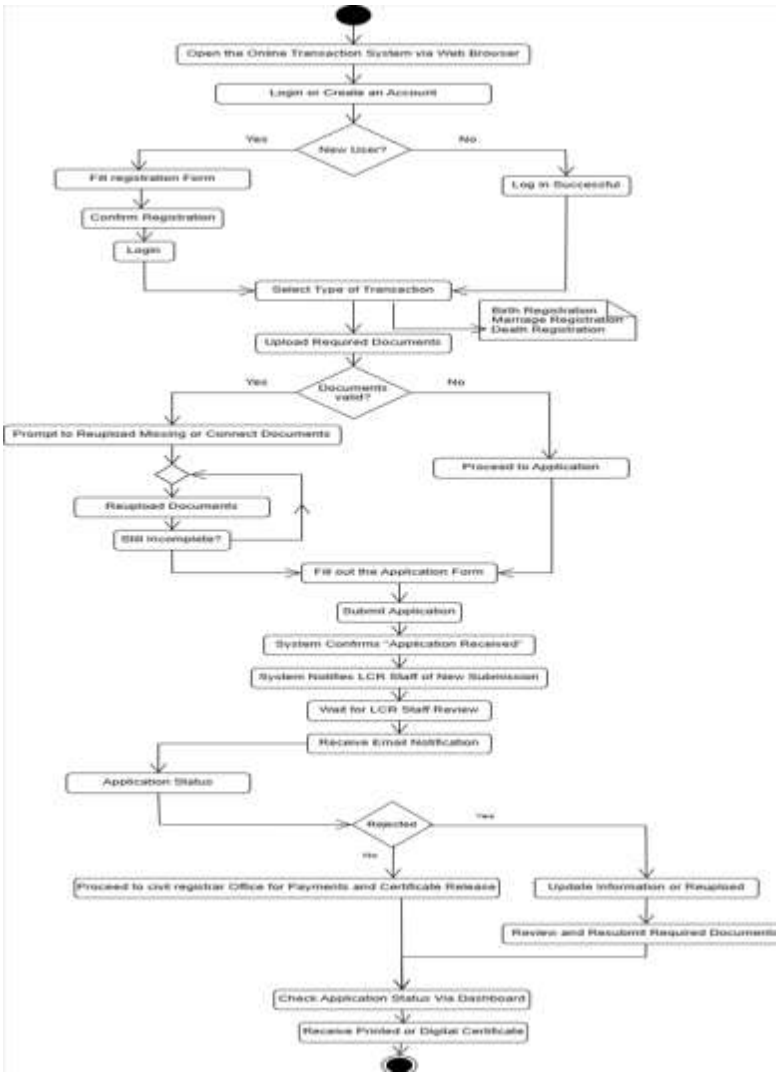



*Figure 6: Activity Diagram Citizen/Applicant Staff*

Figure 6 illustrates the activity diagram flow for an applicant or citizen using the online civil registration system. The process starts with the user opening the online transaction system through a web browser and logging in or registering for an account. If the user is new, they must complete the registration form and validate their registration through email before logging in. After logging in, the applicant selects the type of transaction, i.e., Birth Registration, Marriage Registration, or Death Registration. The system then asks the user to upload documents subsequently. If the documents are incomplete or incorrect, the system asks the applicant to

reupload or edit the documents accordingly. Once all the documents are accurate, the applicant can submit the application form.

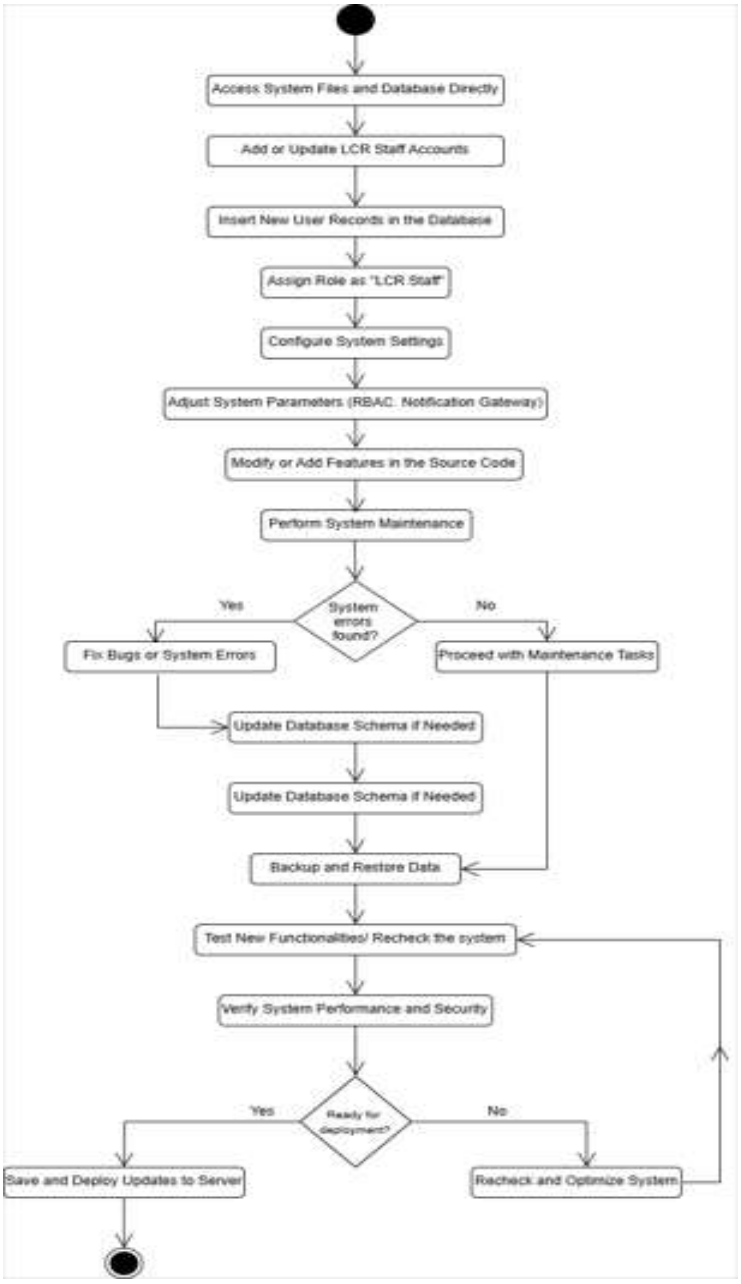



*Figure 7: Activity Diagram Admin/Programmer*

Figure 7 shows the Sequence Diagram of a Programmer or Admin, outlining the technical tasks involved in maintaining, updating, and managing the Local Civil Registry system. The admin

can create or update LCR staff accounts, add new user records to the database, and assign roles such as "Staff associated with the LCR." After that, the admin sets up the system's settings and updates essential settings, such as Role-Based Access Control (RBAC) and the Notification Gateway, to ensure the system operates smoothly and securely.

**2.5 Sequence Diagram**

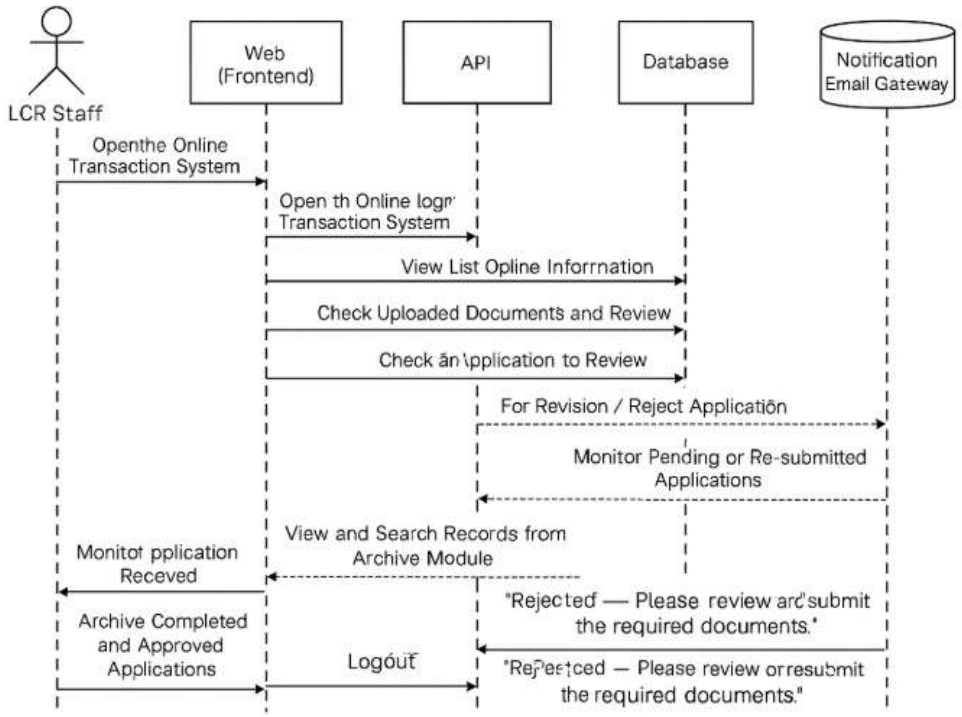



*Figure 8: LCR Staff Sequence Diagram*

Figure 8 illustrates how the LCR staff members work within the system. Staff members register or authenticate via the web interface. Upon authentication, they access a dashboard that presents the total number of registered births categorized by gender. Staff review applications in the processing queue and perform actions, including approval, rejection, or requests for revision. Following each decision, the system communicates the application status to the applicant. This workflow enhances verification, transparency, and accountability in civil registration management.

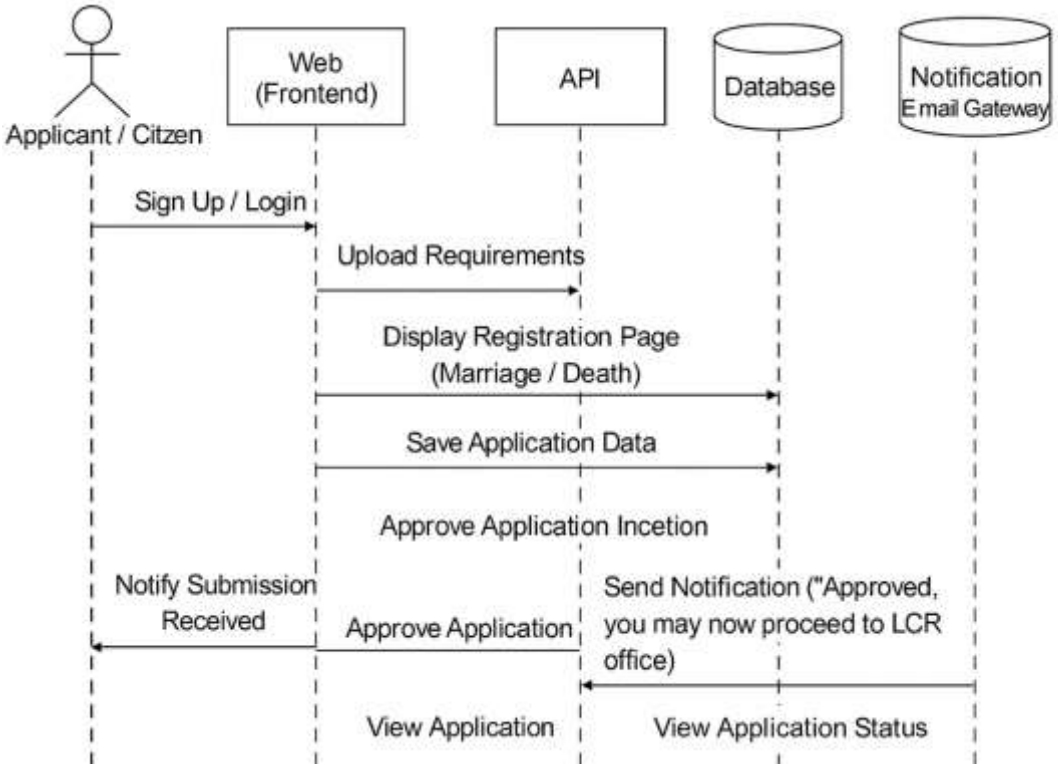



*Figure 9: Applicant/Citizen* Sequence *Diagram*

Figure 9 shows the steps an applicant or citizen follows when using the system. After signing up or logging in through the web interface, users can register or fill out applications for birth, marriage, or death certificates. Depending on the request, the system provides the specific requirements (early or at-early registration for birth or death certificates, or marriage requirements). Upon submission of the application, the system generates an automated confirmation. Following review and approval by administrative personnel, an email containing instructions for visiting the LCR office and completing the process is sent to the applicant. This procedure enhances usability, increases accessibility, and enables efficient application status tracking.

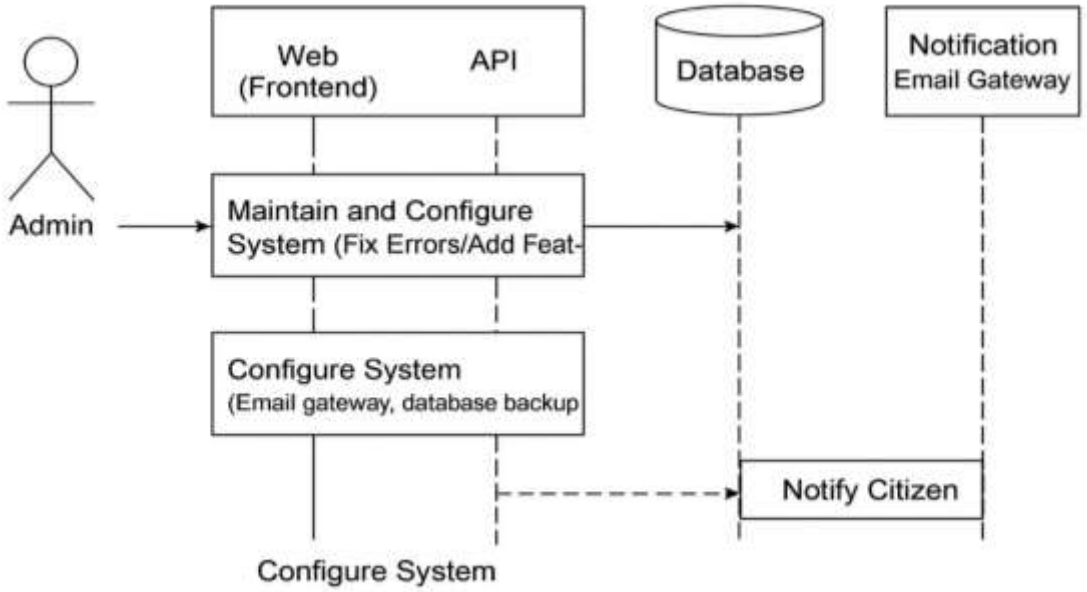



*Figure 10: Admin/Programmer* Sequence *Diagram*

Figure 10 illustrates the actions performed by administrators or programmers within the system. They fix bugs, update software, and add new features to keep things running smoothly. They also set up essential components, such as the email gateway and database backup, to ensure secure communication and data protection. Administrators help keep the system running smoothly and safely by working with the API and database.

**2.6 ERD (Database Design)**

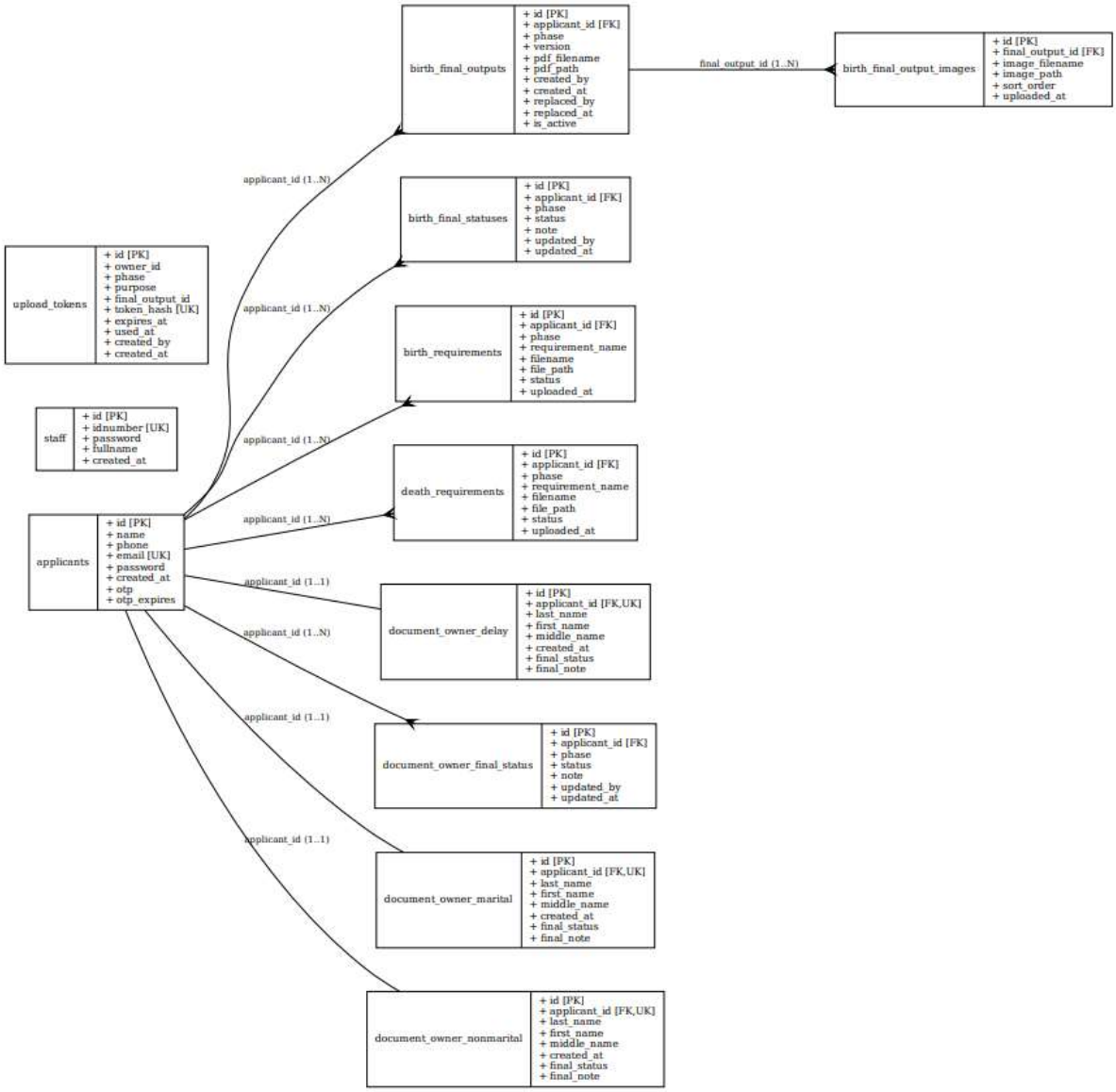



*Figure 11: Entity Relationship Diagram (ERD) Diagram*

Figure 11 illustrates how applicant information, uploaded requirements, application statuses, and final certificate records are related in the database. The diagram highlights the system's structured approach to data management, supporting secure document handling and efficient processing of civil registry transactions.

## 2.7 User Interface Design (Prototype)




*Figure 12: Landing Page Dashboard*

Figure 12 displays the online services interface for the Local Civil Registry Office of Buenavista. The page features options for registering births, deaths, marriages, etc.




*Figure 13: Citizen/Applicant Login Page*

Figure 13 shows the login interface for the Local Civil Registry Office of Buenavista. It allows citizens or applicants to sign in using their email and password to access online services. Additional options include administrator login, account creation for new users, and password recovery for existing users.




*Figure 14: LCR Staff Login Page*

Figure 14 illustrates the login interface for the Local Civil Registry Office of Buenavista. LCR Staff can sign in using their email and password to access online services.




*Figure 15: Account Creation Page*

Figure 15 shows the form that new users must fill out to register for an account at the Buenavista Local Civil Registry Office. To make an account, applicants must give their full name, email address, password, and phone number (optional). This feature enables users to access the system's online services safely.




*Figure 16: Citizen Portal of the Local Civil Registry System*

Figure 16 shows the users' landing page after logging in to the Buenavista Local Civil Registry Office system. The interface provides users with direct links to important services, such as

registering a marriage, birth, or death, and addresses them by name. Users can register for each service on time or late and apply for the necessary documents through the system.



*Figure 17: Birth Registration Requirements Page*

Figure 17 presents the requirements section for death registration at the Local Civil Registry Office of Buenavista. Applicants can select either early/on-time registration or delayed/late registration. The page also provides a detailed list of documentary requirements.




*Figure 18: Death Registration Requirements Page*

Figure 18 illustrates the requirements section for birth registration at the Local Civil Registry Office of Buenavista. Applicants can select either early/on-time registration or delayed/late registration. The page also provides a detailed list of documentary requirements.




*Figure 19: Marriage Registration Requirements Page*

Figure 19 shows the requirements for getting married at the Local Civil Registry Office of Buenavista. The page also provides a comprehensive list of the necessary documents.

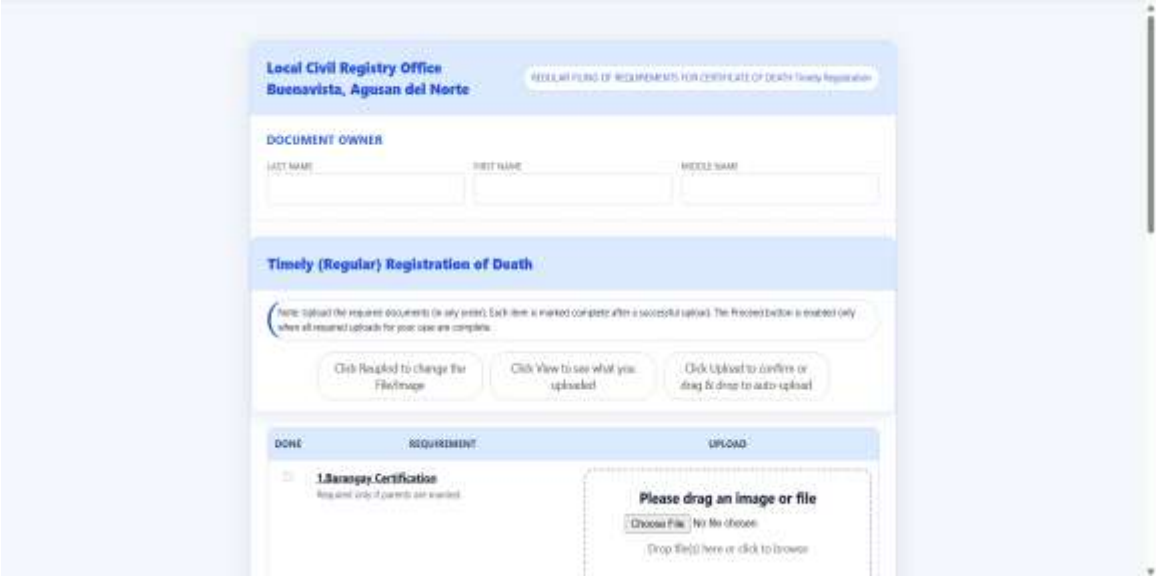



*Figure 20: Registration Requirements Page*

Figure 20 shows the online form used to register births, marriages, and deaths at the Buenavista Local Civil Registry Office. Applicants can upload the required files and input the details of the document owner. To ensure that all requirements, such as the barangay certification and other supporting documents, are submitted before proceeding, the form allows users to reupload, check, and confirm their documents.

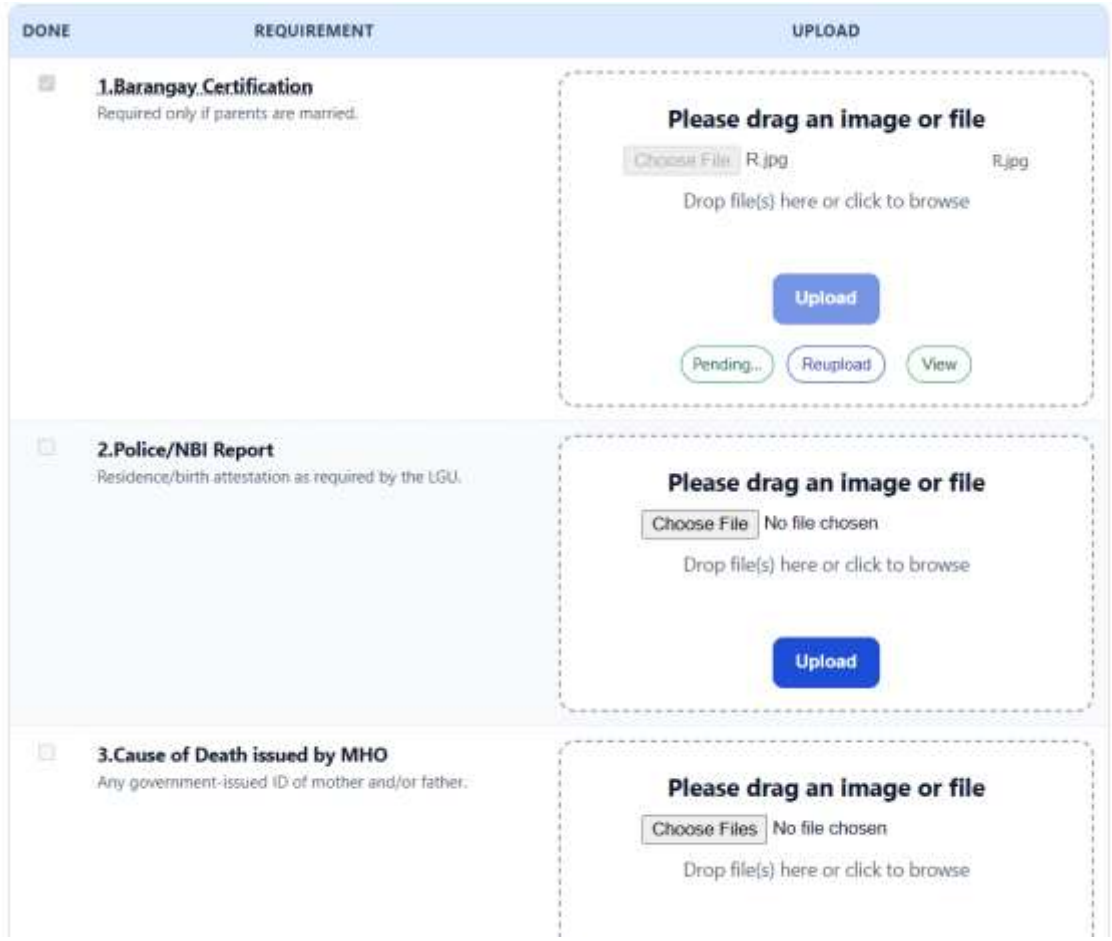



*Figure 21: File Upload Section Registration Requirements*

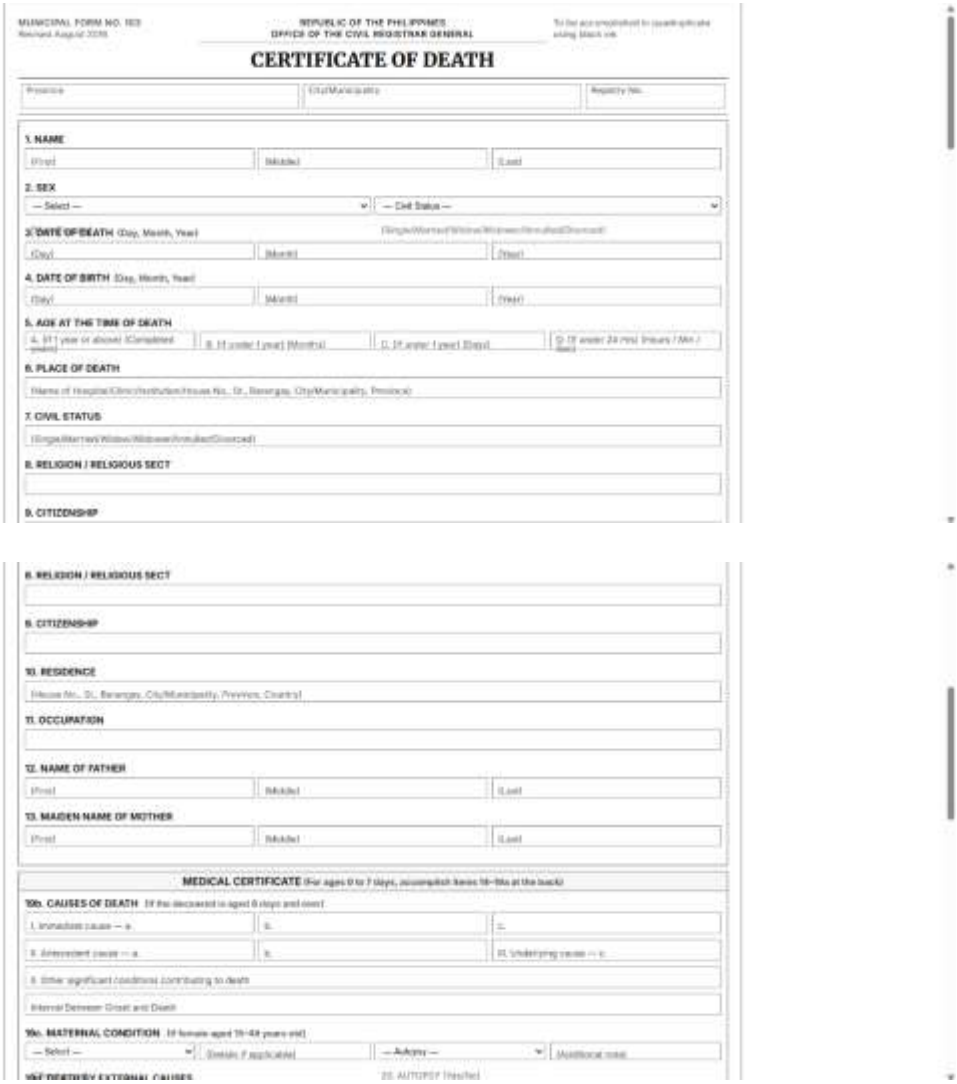
Figure 21 displays the document upload interface for the death, birth, and marriage registration process at the Local Civil Registry Office of Buenavista. It allows users to upload the

necessary documentary requirements. Each requirement section includes options to drag and drop files, upload, reupload, or view the status of the uploaded requirements, such as pending/approved, along with the submitted documents. It provides a convenient and organized way to manage application attachments.




*Figure 22: Certificate of Live Birth Form*

Figure 22 illustrates the Certificate of Live Birth registration form under the Civil Registry System. The LCR staff (Local Civil Registrar staff) review the form after the applicant completes it. The staff can check all the input details before finalizing the record. At the top-right corner of the page, there are two options: Print the form or close the window, allowing staff to print a hard copy of the applicant's birth certificate for official use.



REPUBLIC OF THE PHILIPPINES
OFFICE OF THE CIVIL REGISTRAR GENERAL

**CERTIFICATE OF DEATH**

1. NAME
2. SEX
3. DATE OF DEATH (Day, Month, Year)
4. DATE OF BIRTH (Day, Month, Year)
5. AGE AT THE TIME OF DEATH
6. PLACE OF DEATH
7. CIVIL STATUS
8. RELIGION / RELIGIOUS SECT
9. CITIZENSHIP
10. RESIDENCE
11. OCCUPATION
12. NAME OF FATHER
13. MAIDEN NAME OF MOTHER

MEDICAL CERTIFICATE

10b. CAUSES OF DEATH

10c. MATERNAL CONDITION

*Figure 23: Certificate of Death Form*

Figure 24 shows the digitized version of the Certificate of Death (Municipal Form No. 103, Revised August 2016) employed by the Local Civil Registry Office of Buenavista. This digital format ensures adherence to the standard document requirements of the Philippine Statistics Authority (PSA) for correct and systematic death registration.

MUNICIPAL FORM 90 (FORM NO. 2)
Revised January 2007

REPUBLIC OF THE PHILIPPINES
OFFICE OF THE CIVIL REGISTRAR GENERAL

To be accomplished in quadruplicate using black ink

# APPLICATION FOR MARRIAGE LICENSE

| Province | City/Municipality | Received by | Date of Receipt dd/mm/yyyy | Registry No. |
|---|---|---|---|---|
| | | | | |

| Marriage License No. | Date of Issuance of Marriage License dd/mm/yyyy |
|---|---|
| | |

## GROOM

**THE CIVIL REGISTRAR**

Sir/Madam:

May I apply for a license to contract marriage with _____ and to this effect, being duly sworn, I hereby depose and say that I have all the necessary qualifications and none of the legal disqualifications to contract the said marriage, and that the following data are true and correct to the best of my knowledge and information:

**1. Name of Applicant**
(First) (Middle) (Last)

**2. Date of Birth/ Age**
(Day) (Month) (Year) (Age)

**3. Place of Birth**
(City/Municipality) (Province) (Country)

**4. Sex/ Citizenship**
— Select — (Male/Female) (Citizenship)

**5. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

**6. Religion/ Religious Sect**

**7. Civil Status**

**8. IF PREVIOUSLY MARRIED: How was it dissolved?**

**9. Place where dissolved**
(City/Municipality) (Province) (Country)

**10. Date when dissolved**
(Day) (Month) (Year)

**11. Degree of relationship of contracting parties**

**12. Name of Father**
(First) (Middle) (Last)

**13. Citizenship**

**14. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

**15. Maiden Name of Mother**
(First) (Middle) (Last)

**16. Citizenship**

**17. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

**18. Persons who gave consent or advice**

**19. Relationship**

**20. Citizenship**

**21. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

## BRIDE

**THE CIVIL REGISTRAR**

Sir/Madam:

May I apply for a license to contract marriage with _____ and to this effect, being duly sworn, I hereby depose and say that I have all the necessary qualifications and none of the legal disqualifications to contract the said marriage, and that the following data are true and correct to the best of my knowledge and information:

**1. Name of Applicant**
(First) (Middle) (Last)

**2. Date of Birth/ Age**
(Day) (Month) (Year) (Age)

**3. Place of Birth**
(City/Municipality) (Province) (Country)

**4. Sex/ Citizenship**
— Select — (Male/Female) (Citizenship)

**5. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

**6. Religion/ Religious Sect**

**7. Civil Status**

**8. IF PREVIOUSLY MARRIED: How was it dissolved?**

**9. Place where dissolved**
(City/Municipality) (Province) (Country)

**10. Date when dissolved**
(Day) (Month) (Year)

**11. Degree of relationship of contracting parties**

**12. Name of Father**
(First) (Middle) (Last)

**13. Citizenship**

**14. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

**15. Maiden Name of Mother**
(First) (Middle) (Last)

**16. Citizenship**

**17. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

**18. Persons who gave consent or advice**

**19. Relationship**

**20. Citizenship**

**21. Residence**
(House No., St., Barangay, City/Municipality, Province, Country)

Reset Submit

*Figure 24: Application for Marriage License Form*

Figure 24 illustrates the digital version of the Application for a marriage license used in the Local Civil Registry Office of Buenavista. The organizer divides the form into two sections, one for the groom and one for the bride. Each blank requests personal information, including name, date and place of birth, age, sex, citizenship, and address. It also has areas for the Civil Registrar to verify that applicants meet the requirements for marriage in the Philippines and comply with the law.




*Figure 25: Civil Registry Staff Dashboard*

Figure 25 illustrates the dashboard staff use at the Buenavista Local Civil Registry Office. It helps administrators track and check records of marriages, deaths, and births. It has menus for getting around, an overview of the system and its records, and tools for keeping an eye on applications, checking documents, and ensuring record keeping.




*Figure 26: Statistics Dashboard*

Figure 26 shows the statistics dashboard for the Buenavista Local Civil Registry Office. It shows the number of birth, death, and marriage records, along with a brief description. Staff can easily keep track of trends and compare registration data with the help of visual charts, such as a doughnut chart based on gender and a monthly bar chart.




*Figure 27: Birth Timely Marital Registration*

Figure 27 shows the Birth Timely Marital Registration page of the Civil Registry Management System. This page displays a list of registered applicants along with the corresponding birth details of the child. The information is presented in a table format, including the name of the applicant, date of registration, child's full name, sex, date of birth, and place of birth.




*Figure 27.1: Birth Timely Non-Marital Registration*

Figure 27.1 shows the *Birth Timely Non-Marital Registration* page of the Civil Registry Management System. This page lists registered applicants and the corresponding birth details for children born outside of marriage. The records are presented in a structured table that includes the name of the applicant, date of registration, child's full name, sex, date of birth, and place of birth.




*Figure 27.2: Birth Late/Delay Registration Records*

Figure 27.2 shows the Birth Late/Delay Registration Records page of the Civil Registry Management System. This page displays a tabulated list of applicants who registered births beyond the standard registration period. It provides detailed information for each record, including the name of the applicant, date of registration, child's full name, sex, date of birth, and place of birth.




*Figure 28: Civil Registry Records for Timely Natural Death Registration*

Figure 28 shows the Civil Registry Records for Timely Natural Death Registration page of the Civil Registry Management System. This page presents a structured list of death records that were registered within the prescribed period for natural deaths. It provides essential details for each entry, including the name of the applicant, departed person's name, sex, civil status, date of death, date of birth, and place of death.



*Figure 28.1: Civil Registry Records for Timely Accidental Death Registration*

Figure 28.1 shows the page that displays the list of timely accidental death registrations recorded in the Civil Registry Management System. This page presents death records in a tabular format with essential details such as the name of the applicant, date of registration, deceased person's name, sex, civil status, date of death, date of birth, place of death, religion, citizenship, and residential location.




*Figure 29: Civil Registry Records for Late/ Delay Registration*

Figure 29 shows the page that lists late or delayed death registration records in the Civil Registry Management System. The records are displayed in a structured table containing key information such as the name of the applicant, date of registration, deceased person's name, sex, civil status, date of death, date of birth, place of death, religion, citizenship, and residential location.




*Figure 29.1 Civil Registry Records for marriage Registration*

Figure 29.1 shows the civil registry records page for marriage registration of the Buenavista Local Civil Registry Office. It presents a list of registered marriage records with detailed information about the groom and bride, including their full names, dates of birth, sex, citizenship, residential location, and religion. The page also provides a search function for quick record retrieval, as well as options to view the final document and perform actions such as editing records, allowing staff to efficiently manage and verify marriage registration data.

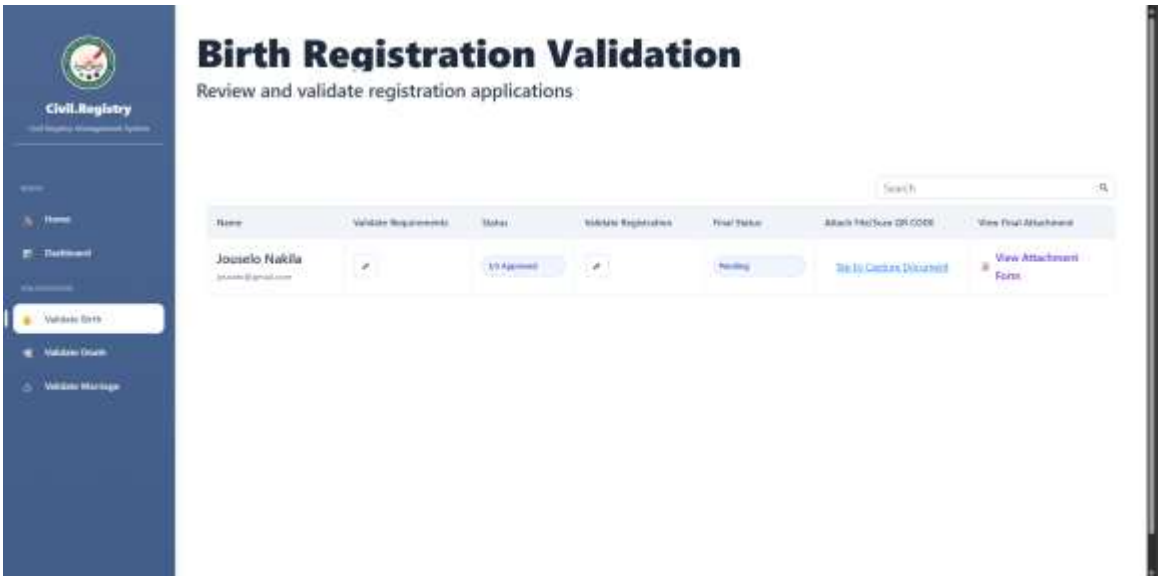



*Figure 30: Birth Registration Validation Interface*

Figure 30 shows the Birth Registration Validation part of the Civil Registry Management System. It allows authorized individuals to review and approve birth registration applications. The interface displays the applicant's name and lets them view the requirements and their registration status. The system displays the current status (Pending) and allows you to review the registration form again.

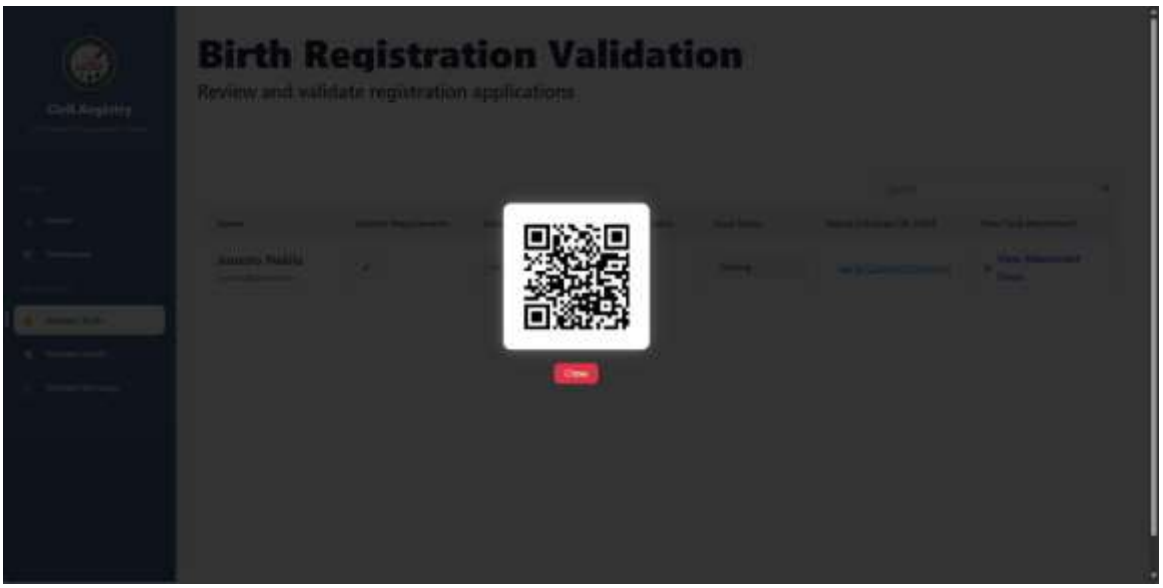


*Figure 30.1: Birth Registration QR code*

Figure 30.1 illustrates the QR code display interface of the Birth Registration Validation module in the Civil Registry Management System. The system generates a unique QR code for each birth registration record. When this QR code is scanned using the camera, it automatically redirects the user to the CS Scanner page or module, allowing staff to verify and capture the corresponding registration document. A “Close” button is provided below the QR code to return to the main validation interface.

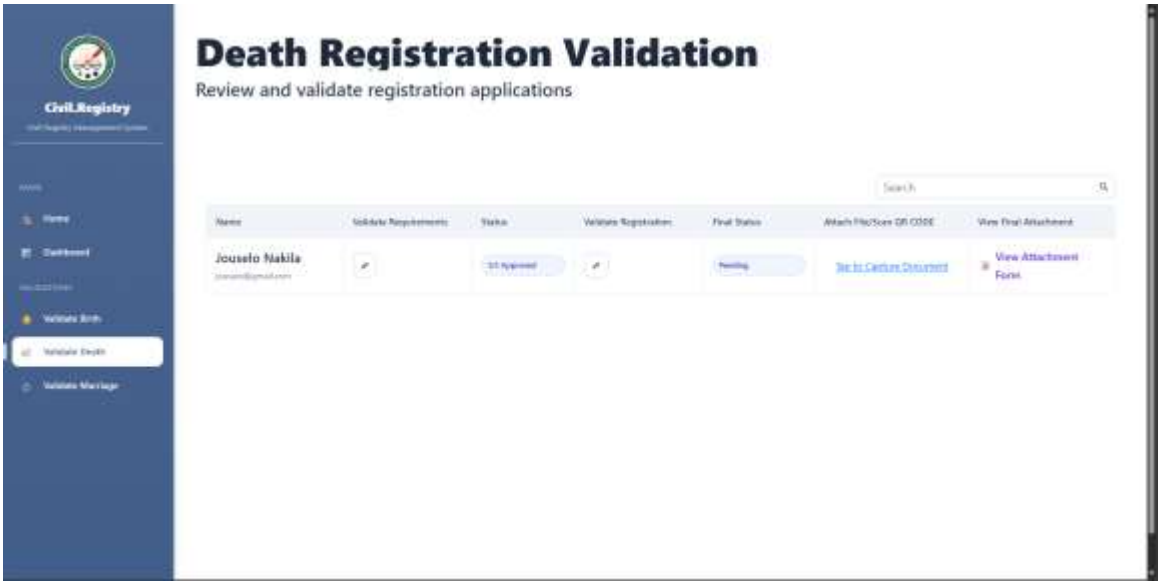



*Figure 31: Death Registration Validation Interface*

Figure 31 shows the Civil Registry Management System Death Registration Validation module. It lets people who have permission check and approve applications for death registration. The interface displays the applicant's name and provides options for checking their requirements and registration status. The system displays the current status (Pending) and allows you to review the registration form again.

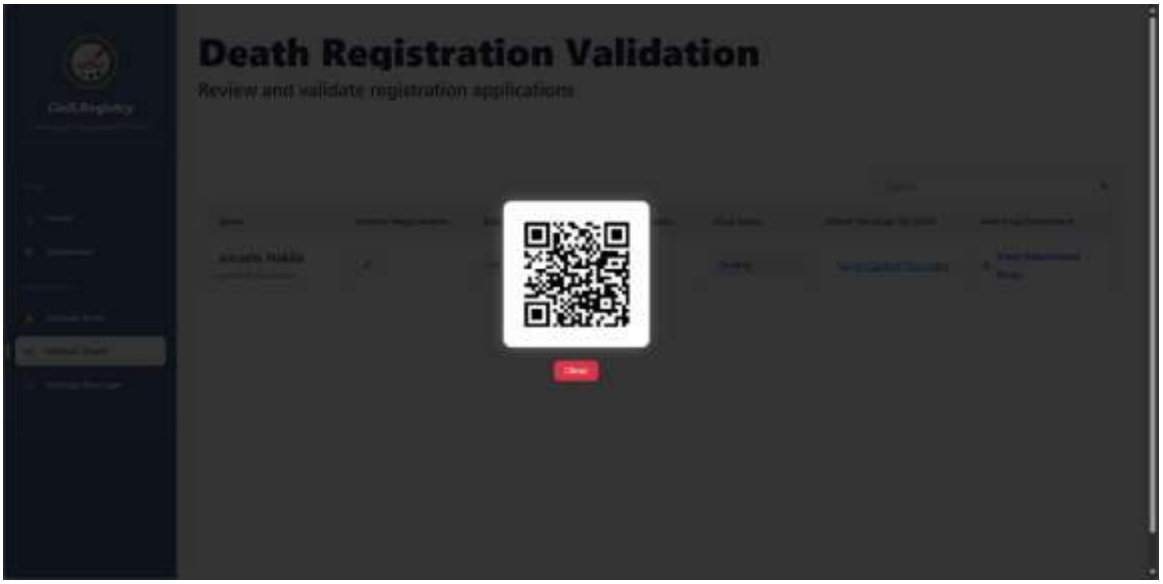



*Figure 31.1: Death Registration QR code*

Figure 31.1 illustrates the QR code display interface of the Death Registration Validation module in the Civil Registry Management System. The system generates a unique QR code for each birth registration record. When this QR code is scanned using the camera, it automatically redirects the user to the CS Scanner page or module, allowing staff to verify and capture the corresponding registration document. A “Close” button is provided below the QR code to return to the main validation interface.

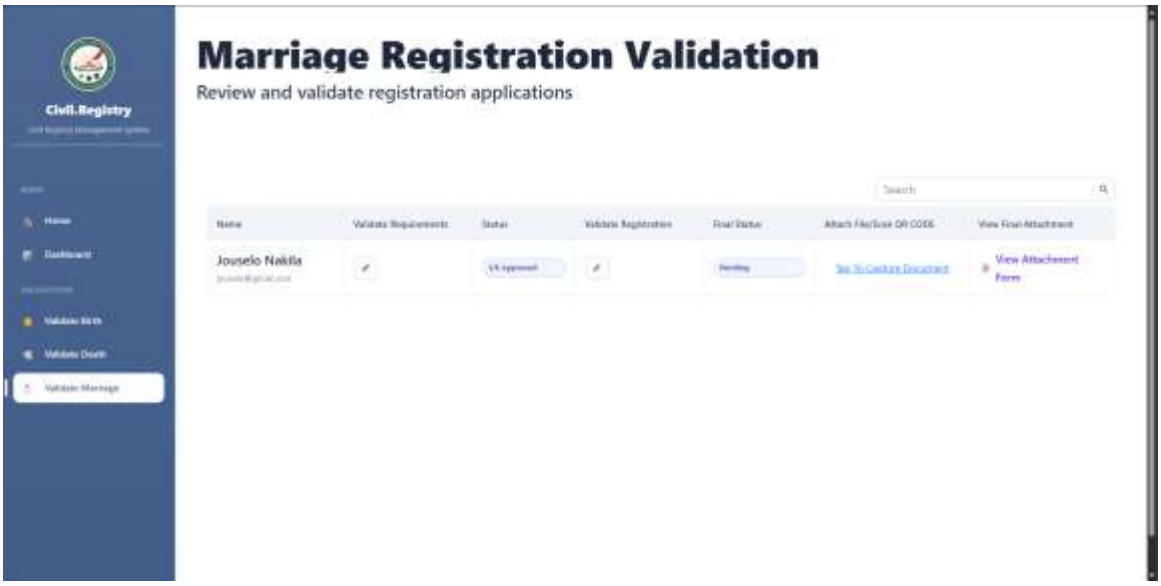



*Figure 32: Marriage Registration Validation Interface*

Figure 32 shows the Marriage Registration Validation part of the Civil Registry Management System. It lets people who can review and approve marriage registration applications. The interface shows the applicant's name and lets them check their requirements and registration status. The system shows the current status (Pending) and lets you look at the registration form again.

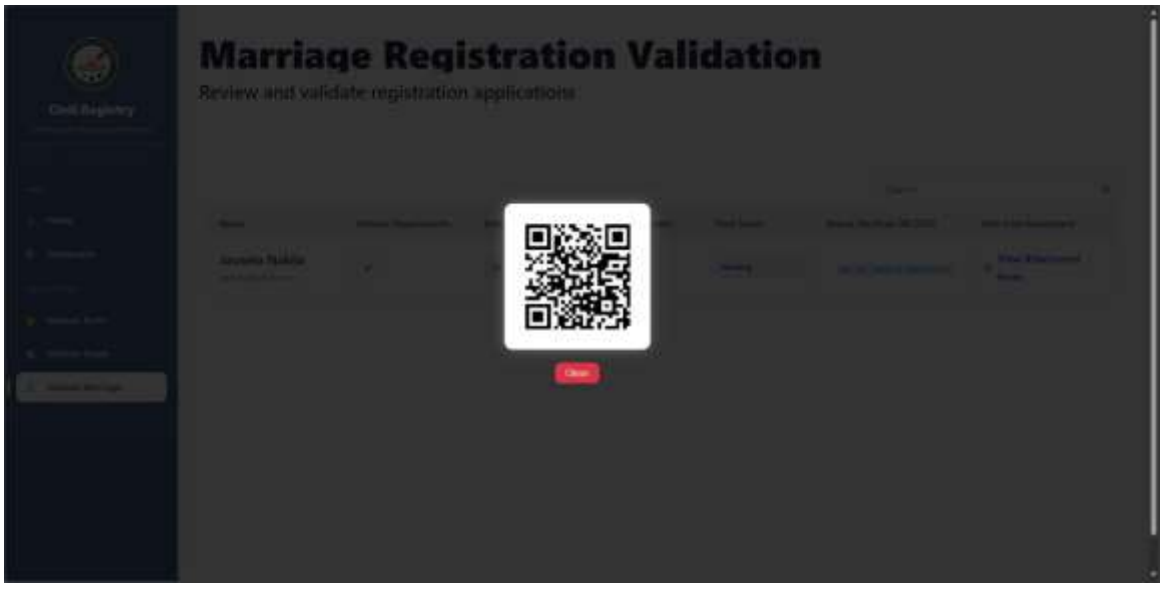


*Figure 32.1: Marriage Registration QR code*

Figure 32.1 illustrates the QR code display interface of the Marriage Registration Validation module in the Civil Registry Management System. The system generates a unique QR code for each birth registration record. When this QR code is scanned using the camera, it automatically redirects the user to the CS Scanner page or module, allowing staff to verify and capture the corresponding registration document. A “Close” button is provided below the QR code to return to the main validation interface.

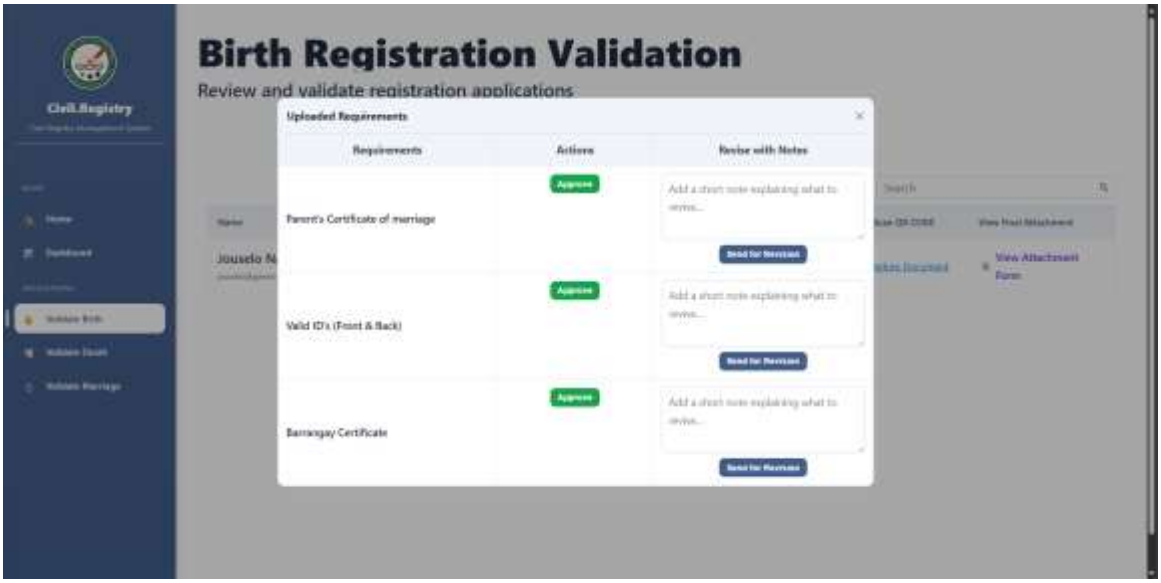



*Figure 33: Uploaded Requirements Interface*

Figure 33 shows the uploaded requirements submitted by applicants, such as birth certificates and parent identification cards. Staff users can approve the document or request revisions by adding explanatory notes. Once a submission is either approved or marked for requirements revision, the applicant is automatically notified through the system and receives an email notification. The email clearly indicates the status of the submission, whether it has been approved or requires further revision and provides any remarks or instructions from the staff. This functionality ensures accuracy, completeness, transparency, and timely communication in the document validation process.




*Figure 34: Certificate of Live Birth Form Interface*

Figure 34 shows the interface for the certificate of live birth, which allows users to input personal and birth-related details directly into the digital version of the municipal form. It supports online data entry, verification, and submission, significantly reducing manual paperwork while maintaining the standardized layout required by the Philippine Statistics Authority (PSA).

### 3.8 Software Platforms, Development Environment and Tools

**Table 1**

**Software Platforms, Development Environments and Tools**

| Components | Specification | Usage |
|---|---|---|
| Internet Browser | Any | A web browser enables users to easily access and display content from the internet. It is responsible for loading the ISO IQA Management System's interface allowing users access to the system's features after a successful login. |
| Front-end | HTML | HTML defines the structural layout of the ISO IQA Management System's interface arranging elements like forms, texts, and other important elements. |
| | CSS | CSS is used for improving the aesthetics and designs of the HTML structure providing a more engaging and responsive website. |
| | JavaScript | To make the website interactive and provide more functionality for the users, JavaScript is used to handle all client-side functionalities. |
| | Bootstrap | Bootstrap allows developers to easily design responsive and mobile-first web pages by providing a collection of pre-designed components, grids, and styles to provide a more responsive website regardless of the screen size and device. |
| | JQuery | jQuery is a quick and efficient JavaScript library that simplifies HTML document manipulation, event handling, |

| | | |
|---|---|---|
| | | animations, and Ajax interactions. It enables developers to write less code for common web tasks and ensures cross-browser compatibility. |
| Back-end | MySQL<br><br>(Database) | MySQL powers data management by providing a robust database, allowing developers to store and manage information efficiently. By utilizing the interface of phpMyAdmin, developers can manage their databases and tables and can also perform queries. |
| | PHP | PHP is used on the server side which manages the interactions between user inputs and the database ensuring smooth data flow. |
| | PHPMailer | PHPMailer is a popular open-source PHP library used to send emails securely and reliably from PHP applications. It provides a simple interface for sending emails via SMTP, PHP's mail() function, or other mail transport methods. |
| Server | Apache | Utilizing the XAMPP control panel, developers can manage the behavioral state of its components ensuring smooth and easy access to local host servers like Apache and MySQL. |

### 3.9 Hardware Requirements

**Table 2**

**Hardware Requirements**

| Components | Specification | Usage |
|---|---|---|
| Device | RAM (16 GB) | RAM facilitates fast and temporary data storage. It is used to enhance the speed and performance of devices in accessing information. |
| | Processor (Intel Core i5) | The processor or the CPU, acts as the central component of the computer, handling the execution of instructions sent from the website. It also facilitates the processing of data. |
| | SSD (128 GB) | Devices employ SSD for storing large amounts of audit files, system software, and reports, while flash storage is commonly used in mobile devices for lightweight data storage. These storage solutions ensure quick retrieval and secure long-term data handling. |
| | Flash Storage (8 GB) | |
| Printer | Any Ink Jet Printer Units | Inkjet printers are utilized for printing audit reports, data summaries, and compliance documents, producing clear and high-quality output. These printers are versatile and affordable, making them suitable for regular printing tasks related to audit management, such as generating official documents for internal use or distribution to stakeholders. |

**Ethical Consideration**

The study entitled *"Online Transaction System for the Civil Registrar's Office of Buenavista, Agusan del Norte"* was conducted in full compliance with institutional research ethics, data privacy regulations, and responsible technology use. The researchers adhered to established ethical principles to protect participants, institutional partners, and sensitive information throughout the system design, development, testing, and documentation phases.

**A. Protection of Intellectual Property Rights (IPR)**

The researchers ensured that this project complies with intellectual property laws and ethical standards. The Online Transaction System for the Civil Registrar's Office of Buenavista was independently developed by the researchers, and its source code, database structure, and system documentation are protected under the Philippine Intellectual Property Code (RA 8293). The official logos of the LGU Buenavista and the Local Civil Registrar's Office (LCR) were used only for system representation, with written permission/authorization secured from the concerned offices, and these remain the property of their respective institutions. The researchers also confirm that the survey questionnaires and evaluation forms used in this study were researcher-made and original, thus protected as part of the research output. All external references and open-source materials were properly credited and used in compliance with their license conditions.

**B. Informed Consent**

Before the system testing and evaluation, the participants, primarily the Local Civil Registrar (LCR) staff and selected citizens, were oriented by the researchers regarding the purpose of the study, the procedures involved, and the potential risks and benefits. The procedures included system orientation, guided task completion, and answering the post-evaluation instrument. The data collected consisted of participants' basic demographic information (e.g., staff

or citizen), usability ratings and questionnaire responses, task-related feedback, and qualitative comments on the system's functionality and user interface. The risks were minimal and limited to possible inconvenience, time spent, or discomfort in answering the evaluation questions, while the benefits included contributing to system improvement and enhancing service efficiency for future users. Written informed consent was obtained prior to participation and was also explained verbally; for participants with limited English comprehension, the briefing was translated into Cebuano to ensure full understanding. The signed informed consent form is attached and referenced in Appendix. No animals were involved in the study; therefore, animal ethics approval was not required.

**C. Data Privacy and Confidentiality**

The study adhered to the Data Privacy Act of 2012 (RA 10173) to ensure the confidentiality and security of all information gathered during system testing and evaluation. The researchers collected only data necessary for research purposes, such as usability ratings, task feedback, and participant comments, and did not store or include citizens' civil registration record contents (e.g., birth, marriage, or death certificate details) in the research dataset. Personally identifiable information was minimized and anonymized through coding prior to analysis, and system-stored sensitive fields such as account passwords and verification credentials were protected using hashing. Access to electronic files and printed consent/evaluation materials was limited to authorized roles only, specifically the researchers and the designated adviser/authorized academic reviewer when required for validation. Anonymized datasets were not shared with external parties (LGU/LCR offices) and were used strictly for academic evaluation; any required review was limited to aggregated or anonymized results. All research data were retained for a defined period after

the completion of the study for documentation and verification purposes and were permanently deleted (digital files) and securely disposed of (printed materials) thereafter.

**D. Voluntary Participation and Freedom to Withdraw**

Participation in this study was entirely voluntary. Prior to system testing and evaluation, the researchers explained the nature and purpose of the study and informed all participants, including LCR staff and selected citizens, that they could refuse to participate or withdraw at any time without penalty or loss of benefits. To prevent workplace hierarchy from influencing participation, recruitment of LCR staff was conducted directly by the researchers and not through supervisors, and no supervisor monitored participation, required attendance, or accessed individual responses. For community respondents, participation was ensured through direct invitation and clear explanation that their decision would not affect any current or future transactions or services from the LGU/LCR. No coercion, incentives, or pressure were used, and all feedback was collected confidentially to support free and honest responses.

**E. Minimization of Harm and Risk Management**

During the system testing and evaluation, the researchers implemented harm minimization and risk management measures to protect participants from emotional and data-related risks. Possible emotional risks included mild frustration, confusion, or discomfort while performing tasks or answering evaluation questions; to address this, participants were given a clear orientation, step-by-step guidance, and were allowed to pause, ask questions, skip any item, or withdraw at any time without penalty. Potential data exposure risks during testing were mitigated by using test or dummy accounts and limiting the collection of personal information to only what was necessary for participation; no civil registration record contents were accessed or recorded for research purposes, and all responses were handled confidentially. Technical support was provided

throughout the session by the researchers, including immediate assistance when participants encountered errors, and contact details were made available for follow-up concerns. These safeguards ensured participant safety, reduced stress during system use, and maintained privacy and security during the evaluation process.

**F. Beneficence and Contribution to Knowledge**

The study contributes to society by supporting improved public service delivery and e-governance through the development of a more efficient Online Transaction System for the Civil Registrar's Office. The direct benefits to participants were limited, as no monetary or service privileges were provided; however, LCR staff directly benefited from exposure to a streamlined workflow and reduced manual processing during testing, while citizens directly benefited from experiencing a more convenient and time-saving transaction process. Indirectly, participants' feedback contributed to improving system usability, reliability, and service quality for future users and the community. The study results and summarized findings will be made accessible to participants upon request through the researchers and in coordination with the Civil Registrar's Office.

**G. Justice and Fair Participant Selection**

Participants were selected using clear inclusion and exclusion criteria to ensure fairness and relevance to the civil registration process. The inclusion criteria were: (1) LCR personnel currently involved in civil registration-related tasks, and (2) citizens of legal age who have prior experience transacting with the LCR office or requesting civil registry documents, and who were willing to participate in system testing and evaluation. The exclusion criteria were: (1) individuals below 18 years old, (2) persons who were not actual or potential users of the system, and (3) individuals who were unable to provide informed consent. Vulnerable groups such as senior citizens and

persons with disabilities were not excluded; they were allowed to participate provided that they met the inclusion criteria and could give informed consent, with assistance or accommodations offered as needed during testing. Representativeness was supported by selecting participants from both major user groups (staff and citizens) and ensuring participation from varied backgrounds and levels of digital familiarity to capture diverse usability feedback and reduce selection bias.

**H. Data Integrity and Accuracy**

To ensure data reliability, the researchers used standardized instruments based on the ISO/IEC 25010 software quality model, specifically a structured system evaluation questionnaire using a Likert-scale rating for the quality characteristics (e.g., functional suitability, usability, reliability, performance efficiency, security, maintainability, and portability), supported by task-based testing and an open-ended feedback form for comments and suggestions. The instruments were validated through expert review by the research adviser/panel and pilot-tested with a small group of users to check clarity, relevance, and consistency, and revisions were applied before final administration. To reduce potential biases, the researchers recognized and controlled for possible sources such as limited sample size, participant familiarity with digital systems, social desirability bias in staff responses, and the restriction of testing to one municipality; anonymity of responses and standardized instructions were applied to encourage honest feedback. AI tools were used only for language and formatting support: ChatGPT was utilized for outlining and improving sentence structure (no data processing, scoring, or interpretation), while Grammarly was used for grammar and readability correction; all AI-assisted suggestions were reviewed and finalized manually by the researchers. The study also acknowledged limitations, including the sample size, the scope and location of testing, and the absence of an online payment feature, which may affect generalizability and system coverage.

**I. Transparency in Reporting**

All findings, results, and analyses were reported truthfully and accurately, with no data manipulation, fabrication, or selective omission. The researchers declare that there were no financial conflicts of interest; any assistance provided by LGU/LCR personnel was limited to coordination, process clarification, and access facilitation for testing, and they had no control over the study outcomes, data scoring, or interpretation. The researchers also disclosed the study's limitations to support honest reporting, including the limited sample size, the conduct of testing within a single municipality, dependence on participant availability and internet connectivity during evaluation, the restricted scope of system features (e.g., no online payment integration), and the possibility of usability bias due to varying levels of digital literacy among participants. All copyrighted, referenced, and acknowledged materials, including external sources and tools used in system development, were properly cited in the manuscript and reference section in accordance with academic standards.

**J. Use of Patented or Copyrighted Materials**

The system utilized only open-source and properly licensed tools and libraries, including PHP and MySQL database services deployed through the XAMPP local server environment (Apache, MariaDB/MySQL, and PHP), as well as the PHPMailer library for email notification functions, which is distributed under an open-source license (LGPL-2.1). No patented technologies were used in the development of the system. All third-party tools, libraries, and documentation were used strictly in accordance with their respective license conditions and were properly acknowledged and cited in the manuscript to avoid copyright infringement. Official logos of the LGU Buenavista and the Local Civil Registrar's Office were used solely for system representation

with written permission/authorization from the concerned offices, and these remain the property of their respective institutions.

**K. Considerations for Human Subjects**

The study involved human participants, specifically LCR staff and community respondents, who were informed prior to participation about the purpose of the study, the procedures to be performed (system orientation, guided task completion, and post-evaluation feedback), and the possible risks and benefits. Risk communication was clearly explained by stating that participation carried only minimal risks such as inconvenience, time spent, or possible confusion/frustration during system use, and that assistance would be available throughout the session; participants were also reminded that they could refuse to answer any item or withdraw at any time without consequences. The participants consisted of 6 individuals in total ( 4 LCR staff and 2 citizens), all within the age range of 50 years old; only adults (18 years old and above) were included, and no minors were involved among the community respondents. Participant welfare, anonymity, and privacy were protected through confidential handling of responses and removal/coding of personal identifiers prior to analysis. No animals were involved in the project; therefore, ethical clearance for animal use was not required.

**L. Responsible Use of AI and Related Technologies**

AI tools were used in a limited and responsible manner to support both manuscript preparation and system development. Specifically, ChatGPT was utilized for concept organization, grammar and sentence refinement, and as a programming support tool for troubleshooting errors and suggesting possible solutions for difficult coding parts (e.g., debugging, syntax guidance, and code optimization suggestions). However, all system requirements analysis, system design decisions, final coding implementation, testing, and deployment were performed by the

researchers, and AI tools were not used to generate research data, compute evaluation scores, interpret results, or write conclusions based on unverified outputs. All AI-generated suggestions for coding and writing were treated as recommendations only and were carefully reviewed, modified, and validated through manual checking, test execution, and comparison with official documentation to ensure correctness, security, and originality. Safeguards against algorithmic bias and inaccuracies were ensured by restricting AI use to supportive tasks, verifying outputs before adoption, and basing all findings and interpretations solely on actual collected data and the researchers' judgment.

**M. Ethical Approval**

The study protocol and research instruments were submitted to the Saint Michael College of Caraga Institutional Research Ethics Committee (IREC) for ethical review for Academic Year 2025–2026. At the time of writing, the ethical clearance is pending, and the researchers will not conduct participant-based testing, data collection, or evaluation activities until the official approval certificate and reference number are issued. Once released, the approval certificate (with date and Certificate/Approval No.) will be attached and cited in Appendix.

**System Evaluation**

This section describes the different evaluation procedures done to the system to assess its functional suitability, performance efficiency, and usability. The survey instrument is adapted from ISO 25010 Software Product Quality Standards; the test scale gives four options to the respondents and will rate the system based on the question with 4 as the highest and 1 as the lowest.

*Data were analyzed and interpreted based on the following parameter*

**Table 3**

**Functional Suitability**

| Adjectival Rating | Scale Range (Mean) | Verbal Interpretation |
|---|---|---|
| 4 | 3.50 – 4.0 | Very Functional |
| 3 | 2.50 – 3.49 | Functional |
| 2 | 1.50 – 2.49 | Moderately Functional |
| 1 | 1.0 – 1.49 | Poor Functional |

Table 3 presents the criteria used to evaluate a product or system's functional suitability according to the ISO survey.

**Table 4**
**Performance Efficiency**

| Adjectival Rating | Scale Range (Mean) | Verbal Interpretation |
|---|---|---|
| 4 | 3.50 – 4.0 | Very Efficient |
| 3 | 2.50 – 3.49 | Efficient |
| 2 | 1.50 – 2.49 | Moderately Efficient |
| 1 | 1.0 – 1.49 | Poor Efficient |

Table 4 outlines the scoring parameters used to assess the system's overall performance efficiency in executing its functions and tasks.

**Table 5**

**Usability**

| Adjectival Rating | Scale Range (Mean) | Verbal Interpretation |
|---|---|---|
| 4 | 3.50 – 4.0 | Very Usable |
| 3 | 2.50 – 3.49 | Usable |
| 2 | 1.50 – 2.49 | Moderately Usable |
| 1 | 1.0 – 1.49 | Poor Usable |

Table 5 presents the scoring parameters used to the usability of the system.

**Table 6**

**Respondent's Distribution**

| SMCC Learning Resource Center | | |
|---|---|---|
| **Position** | **N (Population)** | **Percentage (%)** |
| Personnel | 1 | 12.5% |
| Student | 7 | 87.5% |
| **Total** | **8** | **100** |

Table 6 outlines the distribution of respondents who participated in evaluating the system within the SMCC Learning Resource Center. The respondents are categorized based on

their positions, with 12.5% holding personnel roles and 87.5% being students. This indicates that most feedback comes from students directly interacting with the system. Since they are the primary users, their insights provide a more accurate assessment of the system's functionality, efficiency, and usability in real-world scenarios.

**CHAPTER 4**

**SUMMARY, CONCLUSION, RECOMMENDATION,**

**5.1. Summary of Findings**

## REFERENCES

[1] E. C. Tapangan, *Digital Transformation for Efficiency and Resiliency in Civil Registration*, Philippine Statistics Authority, 2021. Accessed: Aug. 15, 2025. ++++++++++++++++++++++++++++++++++https://psa.gov.ph/system/files/nccrvs/Session%2

012_CCR%20Evangeline%20C.%20Tapangan_Digital%20Transformation%20for%20Efficiency%20and%20Resiliency%20in%20Civil%20Registration.pdf

[2] Ndiyun, R. K., & Mukonza, R. M. (2025). Digital Transformation of Civil Registration System in Cameroon: Innovations in e-Governance. Journal of Digital Technologies and Law, 3(1), 7–34. https://doi.org/10.21202/jdtl.2025.1

[3] S. Naqvi *et al.*, "Post-Pandemic World and Challenges for E-Governance Framework," 2021. http://www.warse.org/IJATCSE/static/pdf/file/ijatcse1561032021.pdf

[4] L. E. A. Villena, R. A. C. Robielos, and W.-J. Shiang, "Analysis on acquisition of Philippine Civil Registry Documents and inclination towards paperless e-government," *IOP Conference Series: Materials Science and Engineering*, vol. 1072, no. 1, p. 012058, 2021. [Online]. Available: https://doi.org/10.1088/1757-899X/1072/1/012058

[5] E. C. Tapangan, "Digital Transformation for Efficiency and Resiliency in Civil Registration," presented at the *1st National Convention on Civil Registration and Vital Statistics*, City Civil Registry Office – Cagayan de Oro, Sep. 1–3, 2021. [Online]. Available: https://services.cagayandeoro.gov.ph/marriagelicense/

[6]R. Angin and P. R. Adawiyah, "Population and Civil Registration Public Services Digital Transformation During the Covid-19 Pandemic," *Journal of Government and Civil Society*, vol. 7, no. 1, pp. 140–158, Apr. 2023 https://doi.org/10.31000/jgcs.v7i1.7180.

[7]J. G. Celeste and M. V. Caelian, "Civil registration awareness, compliance, purpose, and

challenges among municipal residents in the Philippines," *Philippine Social Science Journal*, vol. 4, no. 2, pp. 104–114, Apr.–Jun. 2021. https://doi.org/10.52006/main.v4i2.336>

[8] Y. Vicente, R. Vizarreta, C. Rojas, and M. Ledesma, "Digitalization and satisfaction among Peruvian users towards their civil registration office," *International Journal of Data and Network Science*, vol. 6, no. 3, pp. 1147–1154, 2022. [Online]. Available: https://doi.org/10.5267/j.ijdns.2022.7.005.

[9] Pakaya, H., Deni, S., & Husain, T. (2025). Implementation of Permendagri Number 7 of 2019 Concerning Online Population Administration Services at The Population and Civil Registration Office of Ternate City. *International Journal of Social Science and Human Research*, 8(6), 4932-4936. https://doi.org/10.47191/ijsshr/v8-i6-102.

[10] A. A. Raheem, J. G. Adekunle, and Y. M. Ibrahim, "Design and implementation of birth and death registration system," *Journal of Software Engineering and Simulation*, vol. 6, no. 2, pp. 1–11, 2020.https://www.researchgate.net/publication/365978002_Design_and_Implementation_of_Birth_and_Death_Registration_System

[11] Allam, A. A., AbuAli, A. N., Ghabban, F. M., Ameerbakhsh, O., Alfadli, I. M., & Alraddadi, A. S. (2021). Citizens Satisfaction with E-Government Mobile Services and M-Health Application during the COVID-19 Pandemic in Al Madinah Region. *Journal of Service Science and Management*, *14*(06), 636–650. https://doi.org/10.4236/jssm.2021.146040

[12]T. Adair, A. Badr, L. Mikkelsen, J. Hooper, and A. D. Lopez, "Global analysis of birth statistics from civil registration and vital statistics systems," *Bulletin of the World Health Organization*, vol. 101, no. 11, pp. 768–776, Nov. 2023. [Online]. Available: https://doi.org/10.2471/BLT.22.289035

[13]S. K. Mulmi, "Vital registration system in Nepal," *Kalika Journal of Multidisciplinary Studies*, vol. 28, pp. 28–33, Jun. 2021. https://www.nepjol.info/index.php/KJMS/article/view/38046.

[14] I. R. Sumirat, M. Nurwinda, A. Sanusi, and Jamaluddin, "The Problem of Regulation on Using Absolute Legitimacy in Issuing Population Document," *PETITA: Jurnal Kajian Ilmu Hukum dan Syariah*, vol. 9, no. 2, pp. 674–685, 2024. https://doi.org/10.22373/petita.v9i1.465

[15] A. A. Sordo, A. A. Do, M. J. Irwin, and D. J. Muscatello, "Development of a registration interval correction model for enhancing excess all-cause mortality surveillance during the COVID-19 pandemic," International Journal of Epidemiology, vol. 53, no. 6, pp. 1–5, 2024, doi: **https://academic.oup.com/ije/article-pdf/53/6/dyae145/58803791/dyae145.pdf**

[16] U. Nalband, K. Jadhav, P. Tekade, and N. S. Hunnargi, "Civil Registry Using ASP.NET," International Scientific Journal of Engineering and Management (ISJEM), vol. 4, no. 7, Jul. 2025. doi: **https://www.isjem.com/papers/ISJEM04707.pdf**

[17] M. Anityasari and A. F. Dimyati, "Evaluation of Information System Implementation on Civil Registration Service at Surabaya, Indonesia (Case Study: Birth Certificate Service)," in Proceedings of the Second Asia Pacific International Conference on Industrial Engineering and Operations Management (IEOM), Surakarta, Indonesia, Sept. 14–16, 2021, pp. 3671–3680. https://ieomsociety.org/proceedings/2021indonesia/620.pdf

[18] D. Herdianti, S. Batubara, and M. S. Panjaitan, “Design and Construction of a Mobile Web-Based Birth Certificate Registration Application (Case Study: Population and Civil Registry Service of Binjai City),” Journal of Information Technology, Computer Science and Electrical Engineering (JITCSE), vol. 1, no. 3, pp. 114–119, Oct. 2024, doi: https://ysmk.org/journals/index.php/JITCSE/article/download/114/119

[19] Jeshamar C. Villasis, Kenneth Isaiah Ibasco Abante, "Everyone Should Count: Barriers and Opportunities for Birth Registration in the Philippines" United Nations Children’s Fund (UNICEF) Philippines,2016.https://www.unicef.org/philippines/media/556/file/Everyone%20Should%20Count.pdf

[20] O. Enwe, “Digital Identity Ecosystem and Integrated Birth, Marriage and Death Registers,” ResearchGate, Sep. 2023. [Online]. Available: **https://www.researchgate.net/publication/373707263_DIGITAL_IDENTITY_ECOSYSTEM_AND_INTEGRATED_BIRTH_MARRIAGE_AND_DEATH_REGISTERS**

[21] J. B. Langbaum, N. High, J. Nichols, C. Kettenhoven, E. M. Reiman, and P. N. Tariot, “The Alzheimer’s Prevention Registry: A Large Internet-Based Participant Recruitment Registry to Accelerate Referrals to Alzheimer’s-Focused Studies,” Journal of Prevention of Alzheimer’s Disease (JPAD), vol. 7, no. 4, pp. 242–250, 2020, doi: **http://hdl.handle.net/10150/641758**

[22] Peeler et al., “Centralized registry for COVID-19 research recruitment: Design, development,

implementation, and preliminary results," Journal of Clinical and Translational Science, vol. 5, p. e152, pp. 1–7, 2021, doi: https://www.cambridge.org/core/services/aop-cambridge-core/content/view/10.1017/cts.2021.819

[23] M. Reichold, N. Dietzel, C. Chmelirsch, P. L. Kolominsky-Rabas, E. Graessel, and H.-U. Prokosch, "Designing and implementing an IT architecture for a digital multicenter dementia registry: digiDEM Bayern," Applied Clinical Informatics, vol. 12, no. 3, pp. 551–563, 2021, doi: **https://www.thieme-connect.com/products/ejournals/pdf/10.1055/s-0041-1731286.pdf**

## CURRICULUM VITAE



**NAME: JOUSELO REY G. NAKILA**
**ADDRESS: P- 3A ALUBIHID, BUENAVISTA, AGUSAN DEL NORTE,**
**EMAIL: jouselo_nakila@smccnasipit.edu.ph**

**Personal Information**

Sex : Male
Date of birth : July 18 2005
Height : 5'7
Weight : 74kg
Civil Status : Single

**Educational Attainment**

Elementary : F.S Omayana Elementary School
Secondary : Buenavista National High School
Tertiary : Saint Michael College of Caraga

**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| **None** | **None** |

**CURRICULUM VITAE**



**NAME: ANDREA BUSTILLO CABAN**
**ADDRESS: BRGY 5 PUROK 4, BUENAVISTA, AGUSAN DEL NORTE,**
**EMAIL: alexandrea_caban@smccnasipit.edu.ph**

**Personal Information**

Sex : Female
Date of birth : July 15 2005
Height : 5'3
Weight : 42kg
Civil Status : Single

**Educational Attainment**

Elementary : Libertad Elementary School
Secondary : Buenavista National High School
Tertiary : Saint Michael College of Caraga

**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| **None** | **None** |

## CURRICULUM VITAE



**NAME: GERLY GRACE P. OPMAD**
**ADDRESS: DISTRCT 5, TRIANGULO, NASIPIT, AGUSAN DEL NORTE**
**EMAIL: gerlygracepalomaompad@gmail.com**

**Personal Information**

Sex : Female
Date of birth : November 2, 2004
Height : 5'2
Weight : 39kg
Civil Status : Single

**Educational Attainment**

Elementary : Triangulo Elementary School
Secondary : Nasipit National High School
Tertiary : Saint Micahel College of Caraga

**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| **None** | **None** |

## CURRICULUM VITAE



**NAME: RICHARD JOHN BUDLAT**
**ADDRESS: BRGY 6 PUROK LANSONES, BUENAVISTA, AGUSAN DEL NORTE**
**EMAIL: richardjohnlarede@gmail.com**

**Personal Information**

Sex : Male
Date of birth : July 18 2005
Height : 5’7
Weight : 60kg
Civil Status : Single

**Educational Attainment**

Elementary : Tuyabang Bajo Elementary School
Secondary : Misamis Occidental National High School
Tertiary : Saint Michael College of Caraga

**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| **None** | **None** |

## CURRICULUM VITAE



**NAME: VINCE CARTER YAMBA**
**ADDRESS: BRGY 7 PUROK 6, BUENAVISTA, AGUSAN DEL NORTE,**
**EMAIL: vincecarteryamba@gmail.com**

**Personal Information**

Sex : Male
Date of birth : November 6, 2003
Height : 5'7
Weight : 55kg
Civil Status : Single

**Educational Attainment**

Elementary : Buenavista SPED Elementary School
Secondary : Saint James High School
Tertiary : Saint Micahel College of Caraga

**Membership/Affiliations**

| ORGANIZATION | POSITIONS HELD |
|---|---|
| **None** | **None** |

## APPENDICES

A - Letter of Approval with Certificate of Appearance and Photo
B - Instrument with Informed Consent
C - Map of the Research Locale *(Include Coordinates)*
D - Curriculum Vitae
E - Narrative & Photo Documentation
F- Ethics Committee Approval

## CERTIFICATE OF AUTHENTIC AUTHORSHIP

This is to certify that I/we, the undersigned author(s), have solely and honestly written the entire paper titled: "Click here to enter text." All contents, ideas, data, and analysis presented in this paper are the results of my/our own diligent work and intellectual effort. The paper has been crafted to the best of my/our abilities, in full compliance with academic integrity standards and ethical research practices.

I/We further declare that:

1. This paper is free from any form of plagiarism or copyright infringement.
2. All referenced materials have been properly cited, and due credit has been given to the original authors and sources.
3. The work does not contain any falsified or fabricated data or information.

I/We acknowledge that if any rules or laws related to copyright, plagiarism, or other related legal or ethical standards are violated, Saint Michael College of Caraga shall not be held accountable for my/our actions. Any legal, academic, or ethical consequences arising from such violations shall solely be my/our responsibility.

Signed this
Date: Click here to enter text.
Location: Click here to enter text.

Author(s):

Click here to enter text.
Author

Click here to enter text.
Author

Click here to enter text.
Author

Click here to enter text.
Author